

GENERAL THETA FOUNDATIONS I: CAUSAL EXPERIMENTS, PREDICTIVE QUOTIENTS, AND RESOURCE-AWARE REDUCTION

QIAN QI

ABSTRACT. We prove a controlled common-encoder minimax theorem for finite causal experiments. One encoder is selected before the decision task, while each task controller remains independent of the unknown parameter. The attainable upper risk body is the convex hull of executable deterministic designs under an independent shared seed; its support is given by an exact occupancy recursion. Controlled successor-compatible partitions characterize the feasible encoders. Private randomization has a distinct nonconvex flow description, and an optimal shared scheme can be realized privately by charging at most one additional factor equal to the number of decision rows plus one. The same principle yields a common quadratic variance formula and a risk-sensitive moment duality. We construct the least saturated predictive quotient of a finite instrument and compile rational finite sources with finite description, randomization and execution bounds. A delayed-query Gaussian experiment has causal excess of order $S^{-2/d}$ despite a scalar terminal target whose query-aware checkpoint error is of order S^{-2} . Complete predictive-tree, measure-valued, terminal Gaussian and count-experiment developments are retained with their precise hypotheses.

CONTENTS

1. Introduction	2
1.1. Relation to existing theory	4
1.2. Organization and scope	4
2. The marked decision spectrum	5
2.1. Experiments, tasks, and signatures	5
2.2. Typed simulations and invariance	6
2.3. The terminal decision-complete slice	7
3. Controlled common encoders and the minimax risk body	8
3.1. Finite instruments and the information pattern	8
3.2. The variational theorem	9
3.3. Controlled continuation partitions	10
3.4. Private randomization is a different finite problem	11
4. One variational principle: occupancies, Bayes envelopes and pressure	12
4.1. An exact support-function recursion	12
4.2. The common quadratic task family	13
4.3. Exponential criteria and hidden-phase control	14
5. The least saturated finite quotient	15
6. Finite compilation and several charged resources	17
7. A Gaussian experiment with a genuine temporal penalty	19
8. A prior-independent experiment behind predictive coordinates	21
8.1. The causal factorization	22
9. A necessity theorem for causal realization	23

Date: Ninth revision, September 23, 2026.

2020 Mathematics Subject Classification. Primary 62B05, 94A34; Secondary 60G35, 62C10, 68Q45, 37D35.

Key words and phrases. causal experiment, statistical deficiency, predictive quotient, finite-state realization, Wasserstein entropy, finite precision.

10. Finite-prefix, finite-description realization	25
11. The measure-valued slice: sharp presentation entropy	27
12. A dynamic finite-experiment consequence	29
12.1. A theorem-level downstream arrow	30
Supporting theory and model verification	31
Supporting theory and model verification	31
13. Positive experiments and predictive quotients	31
13.1. Actual histories	31
13.2. The future-test quotient	32
13.3. Attainable Bayes geometry	32
13.4. Normalized response and observation modules	33
14. Causal morphisms and charged approximation	33
15. A structural theorem for predictive trees	35
15.1. Local hypotheses and the finite profile	35
15.2. Balance, dilation and exact closure	36
15.3. Static lower bound and the causal converse	37
16. Two model verifications	38
16.1. Markov-renewal orbit geometry	38
16.2. An infinite hidden shift with noisy continual reports	39
17. Finite presentations of measure-valued filters	40
17.1. A nonlinear physical realization	41
18. Sharp finite-experiment budgets	42
18.1. The cardinality obstruction	42
18.2. Positive second-order reconstruction	43
19. A sharp budget for the physical count-to-contact experiment	44
19.1. The A2 matrices and the scope of the consequence	46
References	47

1. INTRODUCTION

An observation can be compressed only relative to questions and an information pattern. When the question is known before compression, one may retain only its answer. When it arrives later, the same machine must preserve the possibility of answering several incompatible questions. With feedback, the retained information also determines which experiment is performed next. The resulting problem is neither a static covering problem nor a collection of separate control problems.

This paper studies the marked decision spectrum of a causal experiment. Its encoder is fixed before the task, its controller and decoder may know the task but not the unknown parameter, and its persistent state is charged throughout the execution. The spectrum records the attainable risk vectors and their dependence on a resource signature. Our principal result is a controlled common-encoder variational theorem. It constructs the risk body from the finite instrument itself, characterizes its feasible continuation partitions, and identifies its lower support function by an exact occupancy recursion. The maximum over tasks is retained; it is not replaced by one favourable scalarization.

To describe the result, let E be a finite controlled hidden-state instrument on a finite horizon, and let D be a finite family of bounded decision tasks. A row $j \in J$ records a parameter within a minimax task, or the declared prior of a Bayesian task. Put $M = |J|$. A deterministic design ω consists of one common causal encoder together with one observation-admissible controller and decoder for each task. Write $R_j(\omega)$ for its row risk, and b_j for the corresponding task's full-history

optimal baseline. These definitions are made precise in Section 3; in particular, a minimax task has one baseline, not a parameterwise oracle.

Theorem 1.1 (The common-encoder variational principle). *For prescribed finite register widths $\mathbf{S}$, let $\Omega_{\mathbf{S}}$ be the deterministic designs. Under an independent shared seed, averaged before the worst task and parameter are selected,*

$$\Theta_{\text{sh}}(\mathbf{S}; D) = \min_{\lambda \in \Delta(\Omega_{\mathbf{S}})} \max_{j \in J} \left\{ \sum_{\omega} \lambda_{\omega} R_j(\omega) - b_j \right\} \quad (1.1)$$

$$= \max_{w \in \Delta_J} \{V_0(z_0(w)) - w \cdot b\}. \quad (1.2)$$

Here V_0 is the finite prescription recursion (4.2)–(4.3), whose update table is common to all tasks and parameters. Its support values determine the whole attainable upper risk body. Deterministic feasible encoders are exactly the controlled successor-compatible history partitions.

An optimal mixture has at most $M + 1$ designs. Removing the shared seed while retaining its design index in the actual register gives

$$\Theta_{\text{pr}}((M + 1)S; D) \leq \Theta_{\text{sh}}(S; D) \leq \Theta_{\text{pr}}(S; D).$$

Private randomization at unchanged width instead has the nonconvex flow characterization (3.8); it need not give the convex body in (1.1).

The complete proof is given in Theorems 3.1 and 4.1 and Propositions 3.2 and 3.3. The underlying convex minimax and separation arguments are classical. The theorem uses them with a coupled feasible set: the encoder is common across tasks, each controller is common across its parameter rows, and a remembered randomization index is an actual resource when it is not an explicitly permitted shared input. These constraints are indispensable. In Example 3.4, the two-label minimax error of a three-symbol experiment is $1/3$ with a shared seed and $1/2$ without one. Thus replacing private kernels by an unpriced convexification would compute a different spectrum.

The same variational principle has two useful specializations. For a family of Bayesian quadratic tasks it is exactly a maximization over one task weight vector of the least conditional variance of a weighted posterior-mean vector, subject to causal compatibility. This strengthens the single-objective partition formula to the common-encoder maximum criterion. It applies directly to the accepted-cell/failure protocol of the A1 manuscript, with a single update on the observed failure symbol. For exponential criteria it is a duality for moments before taking logarithms, with a multiplicative occupancy recursion. In a hidden-phase example it excludes optimizing separately with knowledge of the phase. These are specializations of one feasible execution body, not independent assumptions that their controllers can be chosen jointly.

We also construct a least likelihood-saturated predictive coordinate for every finite instrument. A backward refinement of posterior vectors, next-report probabilities and successor classes gives its exact minimal widths. For rational input, this construction and its reconstruction kernel are computable by finite arithmetic. It makes the finite sufficiency bridge constructive, while identifying its possible cost: a one-class future-report quotient can require 2^d saturated classes after d parameter-revealing observations. Statistical saturation is therefore not a cost-free completion of a predictive quotient.

Finite synthesis yields control of several resources at once. Theorem 6.1 compiles an optimal finite design into a finite program and a dyadic initial lottery. It charges program length, persistent labels, initialization bits, transient work and bit-serial running time. Its excess differs from the optimum by at most $k2^{-p}$ for $k \leq M + 1$ designs and p lottery bits. Synthesis can be exponential

in the horizon; no multiplicative-energy oracle is used in this finite source theorem. The predictive-tree compiler retained later addresses a different, infinite symbolic input class under its explicit geometric and energy-certification assumptions.

A continuous Gaussian example shows that the same causal constraint has quantitative force beyond finite enumeration. Acquire d noisy scalar coordinates before an independent coordinate query is disclosed. The final target is one scalar parameter, but its optimal causal excess is $S^{-2/d}$, whereas a query-aware static checkpoint has excess S^{-2} . The lower bound uses the common register *before* the query, the conditional-variance identity, and a density-volume bound on the vector of posterior means. This is a temporal penalty, not the terminal dimension argument for an append-only Gaussian vector experiment.

1.1. Relation to existing theory. Comparison of experiments by all bounded decisions is the theory of Blackwell and Le Cam [1, 14, 12]. The terminal deficiency representation included below is stated with its compactness hypotheses and the common encoder outside the task supremum. It does not claim a new unconstrained randomization criterion. Filtered experiment comparison is itself an established subject [15]. Likewise, predictive equivalence and continuation congruences have the classical antecedents [13, 16].

Real-time source coding and decentralized control are particularly close neighbours [21, 3, 4]. Nayyar, Mahajan and Teneketzis [20] derive common-information prescription recursions. In their preprint, Theorem 3 gives the dynamic programme, Corollary 5 treats absent common memory with finite local memory, and Theorem 4 simplifies its information state. Our offline occupancy is in this established designer/common-information tradition. The assertion made here is the coupled minimax risk-body identity, its controlled partition converse, and the priced passage between shared and private randomization. An ordinary designer's dynamic programme, by itself, is not asserted to be new.

The retained quantitative sections use functional and separated-cylinder quantization, symbolic trees, nonlinear filtering and experiment approximation [5, 6, 9, 7, 10]. Their geometric hypotheses remain local to those theorems. The new conditional-variance variational principle includes the finite partition problem and explains the delayed-query Gaussian obstruction. It does not identify Wasserstein law presentation with decision-complete statistical deficiency, which retain different output marks.

1.2. Organization and scope. Section 2 fixes the marked spectrum and its transport and terminal comparison properties. Sections 3 and 4 prove the central variational principle. Sections 5–7 construct saturation, compile finite sources and exhibit the Gaussian temporal penalty. The subsequent sections retain the complete sufficient-statistic, partition, predictive-tree, measure-presentation, Gaussian and count arguments. The companion development additionally retains the earlier mathematical bodies and historical introductions with their original hypotheses; none is silently replaced by a stronger assertion.

The general marked spectrum includes standard Borel experiments, whereas the exact controlled minimax theorem proved here has finite horizon, finite hidden/report/control spaces and finitely many decision rows. The Gaussian example is established separately by its displayed continuous argument. In particular, the factor $M + 1$ is not a uniform statement about a growing or infinite task family. A finite-horizon exponential moment formula is not an LDP, an unbounded-operator theorem, or a proof of a microscopic phase decomposition. Those further structures require their own model theorems. The present foundation supplies a controlled common-encoder converse without importing such theorems as unnamed premises.

2. THE MARKED DECISION SPECTRUM

We first fix the object whose finite presentations are to be compared. This requires specifying both the decision tasks and the information available to the decoder. These specifications cannot be recovered from a dimension or a cardinality alone.

2.1. Experiments, tasks, and signatures. An experiment is a standard Borel causal instrument, with an initial law and transition–observation kernels indexed by a parameter θ . Actions, when present, are chosen before the corresponding transition. The same observation-based policy is used at every value of θ . A *mark* d specifies a horizon, decision times, action spaces, a loss in $[0, 1]$, and a risk functional. The latter is either expectation under a declared prior or the supremum over a declared parameter set, taken after selecting one decoder. A minimax decoder does not know the unknown parameter. A mark may instead specify a fixed Bayesian prediction problem with an integrable squared loss; assertions involving total variation below apply only after a uniform loss bound is supplied. When a loss concerns a latent target, the marked experiment includes that target, and a simulation must preserve its joint law with the observations. Equality of report laws alone does not suffice for such a mark.

An encoder Q is an initialized causal transducer

$$I_t \sim q_t(\cdot \mid I_{t-1}, Y_t, A_{t-1}), \quad I_t \in \{1, \dots, S\}. \quad (2.1)$$

After an update, a decoder sees only (t, I_t) and any explicitly declared current side input. It does not see the discarded observation, an output transcript, old actions, or a data-dependent random seed. All persistent decoder state belongs to I_t . Fresh randomization is allowed. A public seed, when allowed, is independent of the experiment; a data-dependent position or modification of that seed is charged. A terminal reconstruction kernel may have a continuous output even when the physical statistic has a finite alphabet.

We use the resource signature

$$b = (S, A, L, P, p, W, \tau; R). \quad (2.2)$$

Here S counts persistent states; A counts physical terminal labels; L bounds raw symbols read per acquisition; P bounds fixed description bits; p is numerical precision; W is transient work space; τ is update time; and R records the permitted independent randomization. A coordinate ∞ means that no bound is asserted. For a precision constraint, the order is the order on permitted implementations, not the numerical order on an accuracy tolerance. The *cardinality signature* constrains only S (and A when explicitly stated). In particular, exact real constants in a fixed program are free only under that signature. All theorems below state which signature they use.

For an encoder admissible at b , let $r_E(Q, d)$ be the infimal value of the declared risk functional over its allowed decoders. Let $\beta_E(d)$ be the infimal risk using the full observation history under the same decision timing. The encoder is selected before the mark; its decoder may depend on the mark. For a set $\mathcal{D}$ of marks define

$$\mathcal{A}_E(b) = \{u \in [0, 1]^{\mathcal{D}} : \text{some admissible } Q \text{ has } r_E(Q, d) \leq u_d \ (d \in \mathcal{D})\}, \quad (2.3)$$

$$\Theta_E(b; \mathcal{D}) = \inf_Q \sup_{d \in \mathcal{D}} \{r_E(Q, d) - \beta_E(d)\}. \quad (2.4)$$

Infima need not be attained. We use the displayed attainable sets, not their product-topology closure: for an infinite task family, taking that closure before a uniform task supremum can change the value. The common encoder and the order of quantifiers in (2.4) are part of the invariant. The collection of attainable sets, baselines, and resource signatures is the *marked decision spectrum*, denoted $\Theta(E)$. This is the meaning of the symbol Θ in the present paper. It is a profile, not an additional physical parameter. Unbounded squared marks are treated separately without restricting the vectors to $[0, 1]$.

Example 2.1 (The encoder cannot follow the task). Let $X = (X_1, X_2)$ consist of two independent fair bits. A terminal statistic has two values. The two marks ask, respectively, for X_1 and X_2 , with zero-one loss. Each mark separately admits risk zero, but one common encoder has worst risk at least $1/4$. Indeed, for each fixed independent coding seed and each encoder label, the two optimal answers form a word in $\{0, 1\}^2$. Two reconstruction words have average Hamming distance at least $1/2$ from the four equally likely input words: two words may be exact, and each remaining word is at distance at least one. Randomization does not lower the average of the two risks. Hence their maximum is at least $1/4$. Conversely, with a public fair coin, reveal one of the two coordinates and guess the other. Thus the value is $1/4$ when that coin is permitted, whereas interchanging $\inf_Q$ and the task supremum gives zero. No such interchange is made in this paper.

2.2. Typed simulations and invariance. A certified simulation $S : E \Rightarrow F$ consists of initialized causal kernels, a monotone resource transformer ρ_S , and an error bound $\epsilon_S(T)$. The kernels do not know θ or the latent variables. For every target-side admissible policy and every parameter, the simulated marked transcript through horizon T has total variation distance at most $\epsilon_S(T)$ from its law in F . Moreover, composing the simulation with any target encoder of signature b produces a source encoder of signature at most $\rho_S(b)$. These are uniform requirements on one simulator, not permission to choose a different simulator for each task or policy. When marks have different horizons, use the corresponding error for each coordinate; a common ϵ below denotes their supremum.

Two certified simulators represent the same morphism when they have the same transformer and error certificate and induce the same marked execution laws under every admissible continuation. Internal register names are immaterial. We use standard Borel kernels modulo this execution equivalence. The exact morphisms have error zero. A K -state cardinality simulator has the transformer $S \mapsto KS$; its own output is not a second free memory register.

Theorem 2.2 (Spectrum transport and resource equivalence). *For bounded marks and signatures as in (2.2), certified simulations have identities and associative composition up to execution equivalence. If $S : E \Rightarrow F$ and $T : F \Rightarrow G$, the composite has transformer $\rho_S \circ \rho_T$ and error at most $\min\{1, \epsilon_S + \epsilon_T\}$. For every target-admissible Q there is a source-admissible Q' with*

$$r_E(Q', d) \leq r_F(Q, d) + \epsilon_S, \quad \beta_E(d) \leq \beta_F(d) + \epsilon_S. \quad (2.5)$$

If a reverse simulation $R : F \Rightarrow E$ is also supplied, then

$$\Theta_E(\rho_S(b); \mathcal{D}) \leq \Theta_F(b; \mathcal{D}) + \epsilon_S + \epsilon_R. \quad (2.6)$$

In particular, mutually exact finite-state simulations give equal ideal baselines and comparable spectra under the two budget dilations. Whenever a positive power exponent of either cardinality profile exists, the other has the same exponent. These assertions also hold coordinatewise for the attainable risk sets, before taking the task supremum.

Proof. Compose the kernels on the product of the two internal states. Fix one policy for G , regard its composition with T as a policy and continuation for F , and apply the certificate for S . Data processing followed by the certificate for T gives the sum of the two errors. The composition of transformers is forced by the same execution. Associativity follows from iterated kernel integration and the canonical reassociation of the triple product of registers; execution equivalence is preserved under composition because its quantifier includes all continuations. The identity kernel, identity transformer, and zero error give identities. This establishes a category for exact simulations and the corresponding error calculus for approximate ones; it is not an assertion of a new general randomization theorem.

Run Q on the simulated reports and use its decoder. Every loss in $[0, 1]$ changes in expectation by at most the variation error. Taking infima over decoders proves the first inequality in (2.5); an arbitrarily good full-history policy proves the second. The reverse simulator implies $\beta_F(d) - \beta_E(d) \leq$

ϵ_R . Subtract the baselines, take the common task supremum, and only then take the infimum over Q . This proves (2.6). No compactness or attainment of an optimal encoder is used.

For exact cardinality equivalence with factors K_1, K_2 , the inequalities are $\Theta_E(K_1 M) \leq \Theta_F(M)$ and $\Theta_F(K_2 M) \leq \Theta_E(M)$. Monotonicity in M and $\log(K_i M)/\log M \rightarrow 1$ sandwich any existing limit of $-\log \Theta(M)/\log M$. The same reasoning applies to a single bounded squared prediction mark; multiplication by its loss bound changes no positive power exponent. $\square$

The theorem is an invariance statement for a *typed* equivalence. Ordinary statistical equivalence without a resource certificate need not preserve a causal implementation budget. Conversely, retaining every emitted label in an external transcript changes the signature and thus changes the spectrum. Products of independent observations and products of memories are different operations: the former concern the experiment, the latter the implementation.

Proposition 2.3 (Forgetting causal and resource structure). *Fix a finite horizon and restrict to uncontrolled instruments and parameter-decision marks. Sending an instrument to its full transcript experiment, and an exact causal simulator to its transcript kernel, defines a functor to parameterized statistical experiments with parameter-independent kernels modulo equality on every parameter law. It preserves identities and composition. It forgets the resource certificate; the resulting statistical equivalence alone does not supply an equality of causal-register spectra.*

Proof. Iterating a simulator's initialized kernels at a fixed input transcript gives a parameter-independent Markov kernel from source transcripts to target transcripts. Iterated integration identifies the transcript kernel of a composite with the composite of these kernels. Execution equivalence implies equality on every source parameter law, so this assignment is well defined on the stated classes. The identity gives the identity transcript kernel. A general inverse transcript kernel need not be causal, and a causal inverse may store a nonfinite state; neither datum is reconstructed by the forgetful operation. $\square$

2.3. The terminal decision-complete slice. In this subsection $E = \{P_\theta : \theta \in K\}$ has standard Borel observation space X , K is compact metric, and $\theta \mapsto P_\theta$ is continuous in total variation. A terminal encoder is any Markov kernel $Q : X \rightarrow [S]$. Only its physical alphabet $A \leq S$ is constrained; transient computation and the reconstruction kernel are unrestricted. Let $\mathcal{D}_{\text{all}}$ consist of all finitely supported priors on K and all finite-action losses in $[0, 1]$. For these families we use $\delta(F, E) = \inf_R \sup_{\theta \in K} \|RF_\theta - P_\theta\|_{\text{TV}}$, where R is a parameter-independent reconstruction kernel and $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$.

Theorem 2.4 (Decision completeness at a fixed common encoder). *For every such Q ,*

$$\sup_{d \in \mathcal{D}_{\text{all}}} \{r_E(Q, d) - \beta_E(d)\} = \delta(QE, E). \quad (2.7)$$

Consequently

$$\Theta_E(S; \mathcal{D}_{\text{all}}) = \inf_{Q: X \rightarrow [S]} \delta(QE, E). \quad (2.8)$$

This is the Blackwell–Le Cam randomization criterion in a compact setting, with the common finite encoder kept outside the decision supremum. In particular, no new priority is claimed for the unconstrained criterion [1, 14, 12].

Proof. A reconstruction R with variation error ϵ transfers each bounded decision rule, proving that the left side is at most the right side. First suppose that both K and X are finite. Put $F = QE$. Finite-dimensional convex minimax applied to the row-stochastic matrices R and the total-variation dual gives

$$\delta(F, E) = \sup_{\pi, h} \left[\sum_{\theta, x} \pi_\theta h_{\theta x} P_\theta(x) - \sup_R \sum_{\theta, i, x} \pi_\theta F_\theta(i) R(i, x) h_{\theta x} \right],$$

where π is a prior and $0 \leq h_{\theta x} \leq 1$. To justify convexity of the dual domain, write its variables as $u_{\theta x} = \pi_{\theta} h_{\theta x}$ with $0 \leq u_{\theta x} \leq \pi_{\theta}$ and $\sum \pi_{\theta} = 1$. Regard h as a reward with action set X . The first term is the reward of the identity decision in E , bounded above by its optimal reward; the second is the optimal reward in F . Thus every dual expression is at most one of the optimal reward differences. Changing reward to loss proves equality.

Here is the approximation needed for general X, K . A dense sequence (θ_j) and positive weights with sum one give $m = \sum_j w_j P_{\theta_j}$ dominating every P_{θ} ; domination follows by total-variation continuity. The set of densities $\{dP_{\theta}/dm : \theta \in K\}$ is compact in $L^1(m)$. Given $\eta > 0$, choose a finite L^1 net and approximate its elements by simple functions on a common finite partition $\mathcal{C}$ of X . Conditional expectation onto $\mathcal{C}$ is an L^1 contraction, so the cell reconstruction $R_{\mathcal{C}}(\cdot | C) = m(\cdot | C)$ satisfies

$$\sup_{\theta} \|R_{\mathcal{C}} P_{\theta} - P_{\theta}\|_{\text{TV}} \leq \eta$$

after choosing the net and approximations sufficiently accurately. Zero-mass cells are irrelevant. Hence $\delta(F, E) \leq \delta(F, \mathcal{C}E) + \eta$. Now choose a finite parameter net on which both F and $\mathcal{C}E$ vary by at most η in total variation. The supremum reconstruction error on K differs from that on the net by at most 2η , uniformly in the reconstruction. Apply the finite result to this net. Its priors are allowed priors on K , and observing X is at least as informative as observing $\mathcal{C}(X)$, so the resulting decision differences for E dominate those for $\mathcal{C}E$. It follows that the left side of (2.7) is at least $\delta(F, E) - 3\eta$. Let $\eta \downarrow 0$. Equation (2.8) now follows by taking the infimum over Q , with no exchange of quantifiers. $\square$

3. CONTROLLED COMMON ENCODERS AND THE MINIMAX RISK BODY

The causal constraint is a constraint on the encoder, not on each task separately. We now give a variational characterization with this constraint inside the feasible set. The distinction between a shared random seed and private fresh randomization is essential to the statement.

3.1. Finite instruments and the information pattern. Fix a horizon T , finite parameter set K , finite hidden spaces Z_t , report alphabets Y_t , and a finite control alphabet A . The instrument is specified by probability vectors ν_{θ} and stochastic matrices

$$K_{\theta,t}(z', y | z, a), \quad 0 \leq t < T.$$

The hidden variable is not available to either the encoder or the controller. A finite collection D of tasks has finite terminal action sets B_d and losses $\ell_{d\theta}(z, b) \in [0, 1]$. Any bounded loss on a finite trajectory, including losses at several decision times, is of this form after adjoining that trajectory to the hidden state. This enlargement is a description of the experiment, not memory supplied to its controller.

The encoder has widths $\mathbf{S} = (S_0, \dots, S_T)$, with $S_0 \geq 1$. A parameter-independent initialization law chooses the initial label before the first control; this label is visible to the controller and is charged in S_0 . Deterministic initialization can be labelled 1 by relabelling states. Before transition t , the task- d controller chooses a from its current label i . The encoder then observes (a, y) and updates the label. At the last time, a task- d terminal rule chooses b . The encoder's update table is the *same* for all tasks and parameters; the controller's tables may depend on d but not on θ . The current control is an input to the ensuing update, not a retained action history. Task-dependent restrictions on legal controls may be imposed on the controller tables. The controller has no additional persistent register.

For a minimax task retain one row $j = (d, \theta)$ per parameter. For a Bayesian task retain one row j with its specified prior κ_j on K . More generally a row has a task $d(j)$ and a probability vector κ_j ; point masses give the minimax rows. Let J be this finite row set, $M = |J|$, and let β_d be the

full-history optimal risk for the *whole* task d . In particular, the baseline of a minimax task is not a separate parameterwise optimum. Write $b_j = \beta_{d(j)}$.

A *deterministic design* ω consists of one family of updates $f_t(i, a, y) \in [S_{t+1}]$, together with all controller tables $a_t^d(i) \in A$ and terminal tables $b^d(i) \in B_d$. Let $\Omega_{\mathbf{S}}$ be the finite set of these designs. Its raw risk vector is denoted $R(\omega) = (R_j(\omega))_{j \in J}$, where

$$R_j(\omega) = \sum_{\theta} \kappa_j(\theta) \mathbb{E}_{\theta}^{\omega, d(j)} \ell_{d(j)\theta}(Z_T, b^{d(j)}(I_T)). \quad (3.1)$$

The expectation uses the actual closed-loop law. In particular, controls are not optimized separately inside hidden-parameter components.

There are two randomization conventions. In the *shared-seed* convention an independent variable is available to the encoder and the task controller throughout the run. It is chosen independently of the task and parameter; the risk averages its realization before taking the supremum over tasks or parameters. The adversary does not select a new parameter after seeing that realization. In the *private* convention there is no such retained shared variable. Each declared kernel uses fresh randomness, with conditional independence given its allowed inputs. Any seed remembered across updates is charged in the label. We write Θ_{sh} and Θ_{pr} for these two specified slices of (2.4). The term shared seed here has the precise meaning of the public independent seed permitted in Section 2; it never contains data from the experiment.

3.2. The variational theorem. Define the upper risk body and its lower support function by

$$\mathcal{U}_E(\mathbf{S}; J) = \text{co}\{R(\omega) : \omega \in \Omega_{\mathbf{S}}\} + \mathbb{R}_+^J, \quad (3.2)$$

$$h_E(\mathbf{S}; w) = \min_{\omega \in \Omega_{\mathbf{S}}} w \cdot R(\omega), \quad w \in \Delta_J. \quad (3.3)$$

Here Δ_J is the probability simplex. The body is defined from controlled execution risks, not from a presumed predictive quotient.

Theorem 3.1 (Controlled common-encoder minimax theorem). *For the finite instrument and information pattern just specified:*

- (1) *The shared-seed attainable upper risk set is exactly $\mathcal{U}_E(\mathbf{S}; J)$. The common-encoder excess satisfies*

$$\Theta_{\text{sh}}(\mathbf{S}; D) = \min_{\lambda \in \Delta(\Omega_{\mathbf{S}})} \max_{j \in J} \left\{ \sum_{\omega} \lambda_{\omega} R_j(\omega) - b_j \right\} \quad (3.4)$$

$$= \max_{w \in \Delta_J} \{h_E(\mathbf{S}; w) - w \cdot b\}. \quad (3.5)$$

An optimum uses at most $M + 1$ deterministic designs. Thus the encoder is chosen once, before the task, in both formulae.

- (2) *The functions $h_E(\mathbf{S}; \cdot)$ determine the whole upper risk body. For two matched task/row systems,*

$$\mathcal{U}_F(\mathbf{S}'; J) \subseteq \mathcal{U}_E(\mathbf{S}; J) \iff h_E(\mathbf{S}; w) \leq h_F(\mathbf{S}'; w) \quad (w \in \Delta_J).$$

In particular equality of all these support functions is equivalent to equality of the attainable upper bodies. Relabelling histories, hidden states or machine labels changes neither assertion.

- (3) *For constant width S , cardinality alone being charged,*

$$\Theta_{\text{pr}}((M + 1)S; D) \leq \Theta_{\text{sh}}(S; D) \leq \Theta_{\text{pr}}(S; D). \quad (3.6)$$

The first inequality is implemented by retaining the design index in the actual register, including the permitted randomized initialization. If a different signature forbids randomized initialization, this inequality refers to its explicitly initialization-enlarged signature. For a

fixed finite row family, any existing positive cardinality exponent is consequently the same under these two randomization conventions.

Proof. First include the decoder and controller tables in the design. For one fixed common encoder their choices for distinct tasks are independent optimization variables, whereas the same choice for one task must work for all its parameter rows. Compactness of the finite products of probability simplexes implies attainment. Hence minimizing jointly over the common encoder and all its task rules gives exactly the original order $\inf_Q \max_d (r_E(Q, d) - \beta_d)$; it does not give a decoder indexed by the unknown parameter.

Realize every update and controller kernel by a uniform variable at its specified time, independent of the instrument. For a shared-seed scheme, condition on the entire shared seed and these variables. One obtains a deterministic common update family and one deterministic controller for each task. Although different tasks then generate different observation laws, the frozen maps are defined on *all* their possible inputs. They therefore form one element of $\Omega_{\mathbf{g}}$. Its conditional risk is computed under the corresponding controlled instrument, not under an incorrectly frozen uncontrolled history law. Averaging gives a convex combination of the finite vectors $R(\omega)$.

Conversely sample a design index with probabilities λ , expose that independent index to encoder and controller, and execute its tables. The encoder law is independent of which task will be requested. Its risk vector is exactly $\sum \lambda_\omega R(\omega)$. Taking coordinatewise upper bounds proves (3.2) and (3.4). Finite-dimensional convex minimax gives (3.5). Equivalently, the primal is the linear programme minimizing v subject to $\lambda \geq 0$, $\sum \lambda = 1$, and $\sum \lambda_\omega R_j(\omega) \leq v + b_j$ for every row. Carathéodory's theorem in $\mathbb{R}^M$ gives support at most $M + 1$ without changing the optimum risk vector. This argument freezes a tape to form its *distribution of designs*; it never moves a worst-task supremum through that distribution.

The body in (3.2) is closed and convex. If a point lies outside it, finite-dimensional separation supplies a linear functional whose infimum on the body exceeds its value at the point. The functional must be nonnegative on every coordinate direction, because the body contains every positive ray starting at one of its points. Normalizing its coefficients places it in Δ_J . Its infimum is (3.3). This proves the comparison assertion, including both directions of equality. A relabelling gives a bijection of the same execution laws and of their deterministic designs.

Every private scheme is also a shared-seed scheme that ignores its extra input. Conversely an optimal shared scheme can use $k \leq M + 1$ designs. Sample its index privately before the first observation and retain (r, I_t) , which has at most kS values. Each subsequent update, control and readout uses this actual pair. This has the same law as the shared scheme; no external transcript or hidden seed is used. This proves (3.6). Monotonicity, together with $\log((M + 1)S)/\log S \rightarrow 1$, proves the exponent assertion. The constant $M + 1$ is not uniform over growing or infinite task families. $\square$

The theorem identifies a finite controlled spectrum by one variational functional. It is more than the transport of a supplied simulation: its feasible set is constructed from the instrument and the prescribed memory, and the converse recovers every admissible execution. Convex minimax, separation and finite mixed strategies are classical tools. Their use here does not reassign their priority. The substantive information restriction is that all tasks share f , while all parameters within a task share its controller, and every retained design index is accounted for when the shared-seed convention is removed.

3.3. Controlled continuation partitions. Let $\mathcal{H}_t$ be the full finite tree of action–report histories, including histories null under a particular parameter or task. Histories null under every possible execution can be assigned arbitrary updates. A family of partitions $\mathcal{P}_t$ is controlled successor compatible when

$$h \sim_t h' \implies hay \sim_{t+1} h'ay \quad (a \in A, y \in Y_{t+1}). \quad (3.7)$$

Proposition 3.2 (Intrinsic feasible partitions). *Deterministic common encoders with widths $\mathbf{S}$ are, up to label names and unused states, exactly the partition families satisfying (3.7) and $|\mathcal{P}_t| \leq S_t$. A deterministic design adds, for each task, a control constant on every current cell and a terminal decision constant on every terminal cell. Thus the risk body and both sides of (3.5) can equivalently be defined on controlled continuation partitions with these cellwise rules.*

Proof. Equal machine states have equal successors on the same action and report, proving necessity. Conversely label the cells and send (C, a, y) to the cell containing hay for any $h \in C$. Condition (3.7) makes this independent of h . The cellwise controller rule supplies the legal control, and induction identifies the actual label with the actual history's cell for every parameter and task. Defining the maps on the whole tree ensures that an action absent under one task has not made its update task dependent. $\square$

3.4. Private randomization is a different finite problem. For completeness there is also an exact, nonconvex description without a shared seed. Write $g_t^d(a \mid i)$ for the private controller, $q_t(i' \mid i, a, y)$ for the common private update, and $v^d(b \mid i)$ for its terminal rule. Let q_0 be a parameter-independent initialization law when randomized initialization is permitted. Then

$$\begin{aligned} p_0^{d\theta}(z, i) &= \nu_\theta(z)q_0(i), \\ p_{t+1}^{d\theta}(z', i') &= \sum_{z, i, a, y} p_t^{d\theta}(z, i)g_t^d(a \mid i)K_{\theta, t}(z', y \mid z, a)q_t(i' \mid i, a, y). \end{aligned} \quad (3.8)$$

The risk of row j is

$$\sum_{\theta, z, i, b} \kappa_j(\theta)p_T^{d(j)\theta}(z, i)v^{d(j)}(b \mid i)\ell_{d(j)\theta}(z, b).$$

Proposition 3.3 (Private flow characterization). *Minimizing the largest baseline-subtracted row risk over the probability simplexes in (3.8) gives exactly $\Theta_{\text{pr}}(\mathbf{S}; D)$. The minimum is attained. With rational input this is a finite polynomial optimization problem, not in general the linear programme (3.4).*

Proof. Induction on the actual execution gives the displayed occupancies and risk. Conversely their kernels implement precisely that law with no extra retained variable. The variables form a compact product of finite simplexes, and every row risk is a polynomial in them. Continuity gives attainment. The shared occurrences of q_t cannot be split by task or parameter; doing so would change the information pattern. $\square$

Example 3.4 (A strict cost for hiding the seed). Observe $X = \theta \in \{1, 2, 3\}$ without error, retain two labels, and finally guess θ with zero-one loss under worst-parameter risk. The full-information baseline is zero. With a shared seed, choose uniformly one of the three two-element subsets and encode the two selected symbols without error. The omitted symbol is necessarily wrong. Each parameter is omitted with probability $1/3$, so the risk is $1/3$. For any fixed seed, the sum of the three correct-reconstruction probabilities of a two-label experiment is at most two; averaging proves the matching lower bound.

Without a shared seed let r_1, r_2 be the two decoder distributions on three symbols. Correctness for symbol x is at most $\max\{r_1(x), r_2(x)\}$. If all three correctness probabilities exceeded $1/2$, one of two probability vectors would exceed $1/2$ at two coordinates, an impossibility. Thus the private risk is at least $1/2$. It is attained by assigning symbol 1 to the first label and symbols 2, 3 to the second, whose decoder tosses a fair coin. Therefore

$$\Theta_{\text{sh}}(2) = \frac{1}{3}, \quad \Theta_{\text{pr}}(2) = \frac{1}{2}.$$

The gap is not removed by allowing randomized private encoding. A single unknown parameter is selected before the seed, as required above.

4. ONE VARIATIONAL PRINCIPLE: OCCUPANCIES, BAYES ENVELOPES AND PRESSURE

The support function in Theorem 3.1 can be computed without first constructing a quotient or listing full decision trees. The relevant state for this calculation is an *offline* joint occupancy. It is not an additional observation available during execution.

4.1. An exact support-function recursion. For $w \in \Delta_J$ put

$$z_0^{d\theta}(z, i) = \mathbf{1}_{\{i=1\}} \nu_\theta(z) \sum_{j:d(j)=d} w_j \kappa_j(\theta).$$

For arbitrary nonnegative arrays $z = (z^{d\theta}(x, i))$, a prescription consists of actions $a^d(i)$ and a *single* table $f(i, a, y)$. Define its linear occupancy update by

$$(\mathcal{T}_{a,f} z)^{d\theta}(x', i') = \sum_{x,i,y} z^{d\theta}(x, i) K_{\theta,t}(x', y \mid x, a^d(i)) \mathbf{1}_{\{f(i, a^d(i), y)=i'\}}. \quad (4.1)$$

The time t is understood from the two spaces in this formula. Define

$$V_T(z) = \min_{b^d(i) \in B_d} \sum_{d, \theta, x, i} z^{d\theta}(x, i) \ell_{d\theta}(x, b^d(i)), \quad (4.2)$$

$$V_t(z) = \min_{a,f} V_{t+1}(\mathcal{T}_{a,f} z), \quad t = T-1, \dots, 0. \quad (4.3)$$

Theorem 4.1 (Occupancy representation of the spectrum). *For the finite controlled instrument,*

$$h_E(\mathbf{S}; w) = V_0(z_0(w)), \quad \Theta_{\text{sh}}(\mathbf{S}; D) = \max_{w \in \Delta_J} \{V_0(z_0(w)) - w \cdot b\}. \quad (4.4)$$

Every V_t is a positively homogeneous, concave, piecewise-linear function on its nonnegative occupancy cone. A minimizing sequence of prescriptions compiles into an actual common-encoder design with the specified widths. No realized task index or hidden parameter is supplied to f .

Proof. Fix a deterministic sequence of prescriptions. Induction in (4.1) shows that $z_t^{d\theta}$ is the joint law of (Z_t, I_t) under that task and parameter, multiplied by its initial weight. Its terminal linear cost is therefore $w \cdot R(w)$. At the last time the terminal tables in (4.2) give precisely all legal terminal rules: $b^d(i)$ is common to every θ in task d .

At an earlier time, choosing $a^d(i)$ and one common f determines the entire next occupancy. Conversely every possible first prescription of a deterministic continuation has this form. Backward induction therefore proves that (4.3) minimizes over exactly all legal future designs. This proves the first equality in (4.4); Theorem 3.1 proves the second. For any fixed continuation the cost is linear in the initial occupancy. There are finitely many continuations, so V_t is the minimum of finitely many linear forms. The stated geometric properties follow.

The occupancy is propagated during design, starting at the known array $z_0(w)$. Once a sequence of minimizing tables is chosen, its future occupancies are deterministic functions of that sequence. Only the finite tables and the actual label are used online. In particular, one must not store the occupancy array without charging its size, nor allow a different prescription for the unobserved realized parameter. $\square$

This recursion is a designer's dynamic programme with coupled information constraints. Common-information prescriptions and belief-state dynamic programming have substantial antecedents [20, 21]. In the cited common-information preprint, Corollary 5 already treats absent common memory and finite local memories; its unconditional information state is an offline occupancy. The additional assertion here is the exact equality with the *common-encoder minimax spectrum*, including its dual adversarial weights, controlled continuation constraint and explicit seed cost. The recursion does not assert polynomial complexity: the number of linear pieces and the number of finite designs can be very large.

4.2. The common quadratic task family. There is a particularly transparent intrinsic form when reports are uncontrolled. Let the finite history law $p_t(h)$ be fixed, let U_{dt} be square-integrable Hilbert-valued targets, and let $\alpha_{dt} \geq 0$. Task d has loss $\sum_t \alpha_{dt} \|U_{dt} - \hat{U}_{dt}\|^2$. The decoder's answers do not affect later observations or updates. Set $m_{dt}(h) = \mathbb{E}[U_{dt} \mid H_t = h]$. For a successor-compatible partition family $\mathcal{P}$, define

$$D_d(\mathcal{P}) = \sum_t \alpha_{dt} \sum_{C \in \mathcal{P}_t} \sum_{h \in C} p_t(h) \|m_{dt}(h) - \bar{m}_{dt,C}\|^2, \quad \bar{m}_{dt,C} = \frac{\sum_{h \in C} p_t(h) m_{dt}(h)}{p_t(C)}. \quad (4.5)$$

Zero-mass cells are omitted. Define the single weighted geometry

$$M_{w,t}(h) = (\sqrt{w_d \alpha_{dt}} m_{dt}(h))_{d \in D}$$

to lie in the Hilbert direct sum of the target spaces.

Corollary 4.2 (Common-encoder variance duality). *For a shared independent seed and the declared widths,*

$$\begin{aligned} \Theta_{\text{sh}}(\mathbf{S}; D) &= \min_{\lambda \in \Delta(\mathcal{P})} \max_d \sum_{\mathcal{P}} \lambda_{\mathcal{P}} D_d(\mathcal{P}) \\ &= \max_{w \in \Delta_D} \min_{\mathcal{P}} \sum_t \mathbb{E} \|M_{w,t}(H_t) - \mathbb{E}[M_{w,t}(H_t) \mid \mathcal{P}_t]\|^2. \end{aligned} \quad (4.6)$$

The minima range over one successor-compatible family of width at most S_t at each time. This is an exact characterization of the maximum over tasks, not a statement only about one scalarized loss. Its private version lies between the values given by this formula at S and by its implementation with $(|D| + 1)S$ states.

Proof. Conditional expectation gives the orthogonal decomposition of each squared risk into its full-history Bayes risk and prediction of $m_{dt}(H_t)$. On a deterministic state cell its unique optimal readout is the displayed cell mean. All tasks can use their respective means simultaneously without changing the common encoder. Thus the coordinatewise best risk vector of a deterministic partition is exactly $D(\mathcal{P})$. There are finitely many history partitions, although the target Hilbert spaces need not be finite dimensional.

Conditioning on an independent seed gives a distribution of these partitions; conversely their mixtures are executable under the shared convention. The proof of Theorem 3.1 now applies to this finite vector set. The identity $\sum_d w_d D_d(\mathcal{P}) = \sum_t \mathbb{E} \|M_{w,t} - \mathbb{E}(M_{w,t} \mid \mathcal{P}_t)\|^2$ is the direct-sum norm formula. Storing the mixture index proves the last assertion exactly as in (3.6). $\square$

For $|D| = 1$, (4.6) reduces to the earlier conditional-variance characterization, Theorem 9.1. For several tasks it supplies its missing minimax convexification. The outer maximizing weight is common to every time; selecting a separate favourable weight or encoder at each checkpoint is not permitted. Compatible predictive trees are one class of feasible partitions of this same functional. Their energy estimates in the supporting theory supply quantitative bounds for that class; they are not hypotheses of Theorems 3.1 or 4.1.

Corollary 4.3 (A1: maximum over finite query and checkpoint tasks). *Consider the actual finite-command accepted-cell/failure protocol of Corollary 9.2, with its fixed prior, finite horizon and parameter-independent command laws. Fix any finite family of weighted quadratic query/checkpoint tasks. Its optimal shared-seed common causal encoder is given by (4.6), using the accepted and summed failure likelihoods of that protocol. At most $|D| + 1$ partition designs are needed. Retaining their index gives the same optimum with at most $(|D| + 1)S$ private states.*

Proof. The observation likelihoods and posterior means are derived in Corollary 9.2 from $g_j k_j(\theta)$ on acceptance and $\sum_j (1 - g_j) k_j(\theta)$ on the one failure report. These give one finite history law for all tasks. Insert its posterior query means in (4.5) and apply Corollary 4.2. The unseen failed

raw-cell label is never used in a partition update. Unlike the earlier scalarization, the present formula retains the maximum over the whole prescribed task family. $\square$

Proposition 4.4 (A1 with private randomness at unchanged width). *In the same finite-command protocol, let $p_t(h)$ be its actual history law, and let $m_{dt}(h)$ be the finite vector of posterior query means for task d , with query weights absorbed into that vector. Write $\eta_t^q(i | h)$ for the conditional label law generated recursively by one common private encoder q . The private maximum-over-tasks excess at unchanged width is exactly*

$$\min_{q,v} \max_{d \in D} \sum_{t,h,i} \alpha_{dt} p_t(h) \eta_t^q(i | h) \|m_{dt}(h) - v_{dt}(i)\|^2. \quad (4.7)$$

The readouts can be restricted to the compact boxes containing the weighted query means. This is a finite polynomial optimization problem in the stochastic update rows and readout coordinates, with coefficients given by the actual likelihood integrals. It has a minimum. No finite arithmetic claim is made when those coefficients are arbitrary reals.

Proof. Given a supported history, the label law is a finite sum of products of the encoder's transition probabilities. The same row is used whenever the preceding label and observed command/report agree, in particular on the common failure report. Conditional orthogonality subtracts the full-history Bayes risk, leaving precisely the displayed expression. Conversely every listed stochastic row and readout is an admissible private machine, so both infima are equal. Projecting readouts into the coordinatewise range of the query means cannot increase squared error. The feasible set is compact, and its objective is the maximum of finitely many polynomials. Continuity proves attainment. $\square$

4.3. Exponential criteria and hidden-phase control. The same support calculation has a multiplicative form. Let $\gamma > 0$ and let $C_{d\theta}$ be a bounded cumulative cost on the finite trajectory, including a terminal decision. Put

$$M_j(\omega) = \sum_{\theta} \kappa_j(\theta) \mathbb{E}_{\theta}^{\omega, d(j)} e^{\gamma C_{d(j)\theta}}.$$

For task d the risk is $\gamma^{-1} \log \max_{j:d(j)=d} M_j$; a Bayesian task has its single prior row. Let $\beta_d^{\exp}$ denote its corresponding full-history baseline.

Corollary 4.5 (Risk-sensitive common-encoder duality). *The shared-seed excess under these exponential risks equals*

$$\Theta_{\text{sh}}^{\exp}(\mathbf{S}; D) = \frac{1}{\gamma} \log \left[\max_{w \in \Delta_J} \min_{\omega \in \Omega_{\mathbf{S}}} \sum_j w_j e^{-\gamma \beta_{d(j)}^{\exp}} M_j(\omega) \right]. \quad (4.8)$$

For additive costs, the inner minimum is computed by (4.1)–(4.3) after replacing the transition by $K_{\theta,t} e^{\gamma C_{d\theta,t}}$ and the terminal loss by $e^{\gamma C_{d\theta,T}}$. The initial row weights are multiplied by $e^{-\gamma \beta_d^{\exp}}$. These are nonnegative occupancies, not normalized posterior probabilities.

Proof. For a mixture of designs, the exponential moment is $\sum_{\omega} \lambda_{\omega} M_j(\omega)$. It is *not* the exponential of an average of logarithmic risks. Since logarithm is increasing, subtracting the baseline and maximizing rows gives

$$\frac{1}{\gamma} \log \max_j \left[e^{-\gamma \beta_{d(j)}^{\exp}} \sum_{\omega} \lambda_{\omega} M_j(\omega) \right].$$

Minimize over λ and apply finite minimax to the bracketed positive moment table. This proves (4.8). Expanding the product of transition weights proves the Feynman–Kac occupancy assertion by the same induction as before. Positivity, not normalization, is what that induction requires. $\square$

For example, let an unobserved phase $\theta \in \{0, 1\}$ have probabilities $1/2, 1/2$, provide no report, and charge $C = \mathbf{1}_{\{a \neq \theta\}}$. Every observation-admissible action, randomized or not, has moment $(1 + e^\gamma)/2$ and risk $\gamma^{-1} \log((1 + e^\gamma)/2) > 0$. Optimizing separately within each phase would instead give zero. The single $a^d(i)$ in the occupancy recursion forbids precisely that exchange. This supplies a finite-model resolution of the common-policy issue arising in phase mixtures; it does not presume that a latent phase decomposition of a microscopic system has already been constructed.

5. THE LEAST SATURATED FINITE QUOTIENT

The variational theorem does not require a sufficient statistic. When such a statistic is desired, the finite instrument admits a direct construction. It distinguishes the cost of prediction from the cost of preserving the whole parameter experiment.

Fix the finite instrument of Section 3. For an action-report history $h = (a_0, y_1, \dots, a_{t-1}, y_t)$ define the unnormalized forward vector by

$$\begin{aligned} u_{\theta,0}(z) &= \nu_\theta(z), \\ u_{\theta,t+1}(hay, z') &= \sum_z u_{\theta,t}(h, z) K_{\theta,t}(z', y \mid z, a), \quad L_\theta(h) = \sum_z u_{\theta,t}(h, z). \end{aligned} \quad (5.1)$$

Choose a reference prior $\pi_\theta > 0$. A history is supported if $\bar{L}(h) := \sum_\theta \pi_\theta L_\theta(h) > 0$. On supported histories put

$$\begin{aligned} \lambda_\theta(h) &= \pi_\theta L_\theta(h) / \bar{L}(h), \\ k_t(y \mid h, a) &= \bar{L}(hay) / \bar{L}(h). \end{aligned} \quad (5.2)$$

The likelihoods in these formulae do not contain action-selection probabilities. Under any parameter-independent observation-based policy, those probabilities cancel from the posterior and from likelihood ratios. Thus the formulae apply to every policy on its supported histories, not only to a policy used to explore the history tree.

A history coordinate $s_t(h)$ is called *saturated and predictive* if it determines $\lambda(h)$, determines every vector $k_t(\cdot \mid h, a)$, and has a deterministic update from (s_t, a, y) on supported continuations. At time T only the first condition is imposed. This is the finite, terminal-parameter version of the three conditions in Section 8. Additional latent-target marks require the corresponding conditional kernels as further decorations; parameter sufficiency alone does not imply their sufficiency.

Construct partitions backwards. At level T group histories with the same vector $\lambda(h)$. Having constructed the class map c_{t+1} , let $c_t(h)$ be the class determined by the finite tuple

$$\left(\lambda(h), (k_t(y \mid h, a), c_{t+1}(hay))_{a,y} \right). \quad (5.3)$$

For an unsupported continuation its class entry is one fixed cemetery symbol. Let K_t be the number of classes at level t .

Theorem 5.1 (Constructive minimal saturation). *For every finite controlled instrument the partitions (5.3) form a saturated predictive coordinate. Every other such deterministic history coordinate refines them at each level. Consequently K_t is its least possible width, simultaneously at every level. The partitions, as equivalence relations on supported histories, are independent of the choice of the strictly positive reference prior.*

There is a parameter-independent finite reconstruction kernel $L_t(y \mid c, a, c')$ which, from successive quotient classes and the chosen controls, reconstructs the report stream under every admissible policy. The original and quotient instruments have mutually exact causal simulations. Their additional persistent state is at most $\max_t K_t$ labels, so composing either simulation with an S -state encoder costs at most $(\max_t K_t)S$ states.

If the instrument and prior have rational entries, the classes and reconstruction kernels can be computed by exact finite arithmetic. The calculation is polynomial in the size of the expanded finite history tree and its rational input length. That tree, and the least width K_t , may grow exponentially with the horizon.

Proof. Equality of the keys at level t implies equality of the posterior vectors, of all reference predictive probabilities, and of the class of every common positive-probability successor. Sending (c, a, y) to that successor class therefore defines an update. Unsupported continuations may be assigned arbitrarily. This proves the three required properties.

For minimality, at time T any saturated coordinate separates unequal posterior vectors, so its fibres refine c_T . Suppose the assertion has been proved at $t + 1$. Equal coordinates at time t have equal posterior and predictive vectors. Their supported successors under any fixed (a, y) have equal next coordinates by recursion, and hence equal c_{t+1} by the induction hypothesis. Their complete keys (5.3) coincide. Backward induction proves the claimed simultaneous refinement, rather than just a bound on a terminal statistic.

To check prior independence, first observe that equality of posterior vectors under a positive prior is equivalent to proportionality of the nonnegative likelihood vectors $(L_\theta(h))_\theta$. This relation is independent of the prior. For a fixed parameter with $\lambda_\theta(h) > 0$, the actual next-report probabilities are

$$P_\theta(y \mid h, a) = k_t(y \mid h, a) \frac{\lambda_\theta(hay)}{\lambda_\theta(h)}. \quad (5.4)$$

In an existing class the right side is constant on histories for every supported successor, because the successor class determines its posterior. The set of parameters with positive likelihood is also constant on the class. Under a different positive prior, its new posterior weights are functions of its old posterior weights; averaging these common parameter-specific predictive kernels therefore gives a common new reference predictive kernel. The existing partition is saturated, predictive and recursive for the new prior. Minimality in both directions proves equality of the two partitions.

For reconstruction, write $F_t(c, a, y)$ for the class update and define

$$H_t^0(c' \mid c, a) = \sum_{y: F_t(c, a, y) = c'} k_t(y \mid c, a),$$

$$L_t(y \mid c, a, c') = \frac{\mathbf{1}_{\{F_t(c, a, y) = c'\}} k_t(y \mid c, a)}{H_t^0(c' \mid c, a)} \quad \text{if } H_t^0(c' \mid c, a) > 0.$$

Choose an arbitrary probability vector when the denominator is zero. The posterior at the next class is constant over the summed reports. Equation (5.4) thus factors the actual joint law of report and next class into this same L_t and the parameter-specific class-transition probability. Conditioning further on the next class cancels the likelihood factor. This gives a single parameter-independent report reconstruction, including when parameters have different supports. Histories having zero likelihood under a parameter are not visited under that parameter.

Starting from the initial class, the reconstruction stores the previous class, receives the next class after the actual chosen action, samples its report using L_t , and replaces its stored class. A target policy then chooses its next action from the reconstructed reports. Induction in time proves equality of the resulting action-report laws for every parameter and policy. In the other direction, apply F_t to the original reports. Product registers give the displayed storage bound; neither simulation presumes a cost-free transcript.

Finally, (5.1) enumerates the rational forward vectors on a finite tree. Formulae (5.2) and (5.3) use only finite sums, products, divisions by positive numbers, and equality tests of rational tuples. Class labels can be assigned by sorting those tuples backwards. To see the stated bit-complexity bound, take a common denominator for all rational input entries. Its bit length is at most their total input length. At level t , a forward entry can be represented with the t th power of this

denominator times the initial denominator; sums add at most the bit length of the number of summands at each step. Thus every intermediate integer has bit length polynomial in horizon and input length. Posterior divisions and tuple comparisons preserve a polynomial bound. There are only as many forward and backward steps as nodes and edges of the expanded tree, times its finite hidden and parameter dimensions. This proves effectivity without any oracle for exact real comparisons. $\square$

The theorem constructs the object assumed by the causal factorization in this finite class. Its minimality is among the *specified* likelihood-saturated, raw-report-predictive coordinates. It is not a claim that every exact simulator, or every task-specific lossy encoder, must use these coordinates. The variational principle optimizes the latter directly and may use substantially less memory.

Example 5.2 (Saturation can have an exponential price). Let the parameter range over $\{0, 1\}^d$, and reveal it noiselessly one bit at a time during the first d observations. Thereafter all reports are independent fair bits. At time d all future-report predictions coincide, so the future-report predictive quotient has one class. Nevertheless the posterior on the parameter is a different point mass for each word. Every saturated coordinate at that time has at least 2^d classes, and the construction above has exactly 2^d . Predictive closure alone therefore supplies no controlled-cost parameter saturation. This is a cost calculation in a concrete model, not an assumption that the missing saturation has bounded size.

Example 5.3 (Likelihoods alone need not close prediction). There is one parameter. Let Y_1 be a fair bit and put $Y_2 = Y_1$. The constant coordinate is recursive and contains every parameter likelihood, but its one-step predictive law at time 1 is not a function of that coordinate. The least saturated predictive widths are $(K_0, K_1, K_2) = (1, 2, 1)$. The last quotient class can discard the bit, because reconstruction of Y_2 still uses the previous class. This example separates the hypotheses of the factorization theorem; it does not assert impossibility of other simulators with other charged resources.

6. FINITE COMPILATION AND SEVERAL CHARGED RESOURCES

The finite variational principle also has an effective implementation. This is a separate source of multi-resource control from the energy oracle used in the predictive-tree construction. The price of synthesis is explicit and can be large; it is not moved into a free online oracle.

Use the finite instrument and finite decision alphabets of Section 3, with rational probabilities and losses. Let $S = \max_t S_t$, $Y = \max_t |Y_t|$, $B = \max_d |B_d|$, and $m = |D|$. A table implementation of one deterministic design has description length bounded, up to an absolute coding constant, by

$$\begin{aligned} P_0 = & \sum_{t=0}^{T-1} S_t |A| |Y_{t+1}| \lceil \log_2(S_{t+1} + 1) \rceil \\ & + m \sum_{t=0}^{T-1} S_t \lceil \log_2(|A| + 1) \rceil + m S_T \lceil \log_2(B + 1) \rceil \\ & + O((T + m) \log_2(T + S + |A| + Y + B + 2)). \end{aligned} \quad (6.1)$$

Unused states and forbidden controller actions can simply be retained as redundant table entries. The externally specified time index is the same clock convention as in (2.1). If no external clock is permitted, its $T + 1$ values must also be included in the register.

Theorem 6.1 (Effective finite-source realization). *An optimal shared-seed common-encoder scheme for the rational finite instrument can be synthesized by a finite exact calculation. It can use $k \leq M + 1$ deterministic tables with rational mixing weights. For every integer $p \geq 1$ there is a*

dyadic implementation whose excess is at most

$$\Theta_{\text{sh}}(\mathbf{S}; D) + k 2^{-p}. \quad (6.2)$$

Its fixed description has $O(kP_0 + kp)$ bits. Initialization uses exactly p independent fair bits; no random bits are required subsequently. With an explicitly permitted read-only shared seed its persistent register has width at most S . Without that convention the same scheme has width at most kS , including the stored design index.

Each update reads one supplied finite-alphabet report. On a bit-serial table implementation, transient space can be bounded by $O(p + \log(P_0k + 2) + \log(S + |A| + Y + B + T + m + 2))$ bits and the running time of an update or initialization by $O(kP_0 + kp + p + 1)$ bit operations, after increasing the universal coding constant. No unit-cost access to an arbitrary real constant is required. Reading the physical report itself remains an input cost, rather than a claim about how an underlying continuous measurement was acquired.

Proof. Enumerate the finite deterministic designs. Their risks are rational numbers obtained from finite products and sums of rational transition entries and losses. Full-history baselines can likewise be computed by enumerating full-history policies and solving their finite minimax linear programmes; for a Bayesian task their minimum is already a finite linear optimization. Thus the primal programme (3.4) has rational data and has a rational optimal basic solution. Rational linear algebra and the usual elimination of affine dependence among its risk vectors give a rational optimal mixture of at most $M + 1$ designs. This gives a finite synthesis algorithm, not a bound polynomial in the horizon or the number of reports. Integer determinants of the finite rational constraint matrix give finite bit bounds for all weights, so synthesis does not require a probability oracle.

Let its weights be $\lambda_1, \dots, \lambda_k$. For $r < k$ set $\tilde{\lambda}_r = 2^{-p} \lfloor 2^p \lambda_r \rfloor$ and let the last weight be the nonnegative remainder. Then

$$\|\lambda - \tilde{\lambda}\|_{\text{TV}} = \sum_{r < k} (\lambda_r - \tilde{\lambda}_r) \leq (k - 1) 2^{-p}.$$

For every row, its risk lies in $[0, 1]$, so the difference of its mixed risks is at most this total variation. Taking the maximum and subtracting the unchanged baseline proves (6.2).

A uniform p -bit integer, compared to the cumulative integer weights, selects the dyadic design exactly. Its deterministic tables then implement the scheme. There are k copies of the tables counted in (6.1) and k integer weights of at most $p + 1$ bits, which proves the description bound. In the private implementation the index is part of (r, I_t) and hence multiplies the width by k . In the shared implementation the seed is explicitly allowed throughout the run; its index can instead be reread from that permitted input. A modified or data-dependent seed would not qualify for this exception.

For a conservative machine model, store the immutable description as a sequential binary table with self-delimited addresses. Locate an update, a controller action, or a terminal answer by scanning it. Counters need only logarithmic space in its length and in the finite alphabet sizes. Comparing the p -bit initialization integer with its cumulative weights requires $O(p)$ extra space and $O(kp)$ time by a sequential scan. Copying the selected finite answer and updating the label has the same bounds. Thus the displayed time and space bounds are valid without a random-access assumption. They are deliberately not optimized. $\square$

Proposition 6.2 (Stability of finite synthesis). *Suppose two finite instruments have matched alphabets, losses and timing. Their initial hidden laws have total variation distance at most η_0 for every parameter, and the corresponding transition–report rows have distance at most η_t uniformly in hidden state and control. Then, for every admissible common-encoder/controller design, the difference of each bounded-loss risk is at most $\varepsilon = \min\{1, \eta_0 + \sum_{t < T} \eta_t\}$. Their full-history baselines*

differ by at most ε , and their shared or private excess spectra at the same signature differ by at most 2ε . An additional uniform error ζ in the loss adds 2ζ to this last bound.

Proof. Couple the initial hidden states maximally. Until the two trajectories first disagree, choose the same controls, report-driven updates and random bits in both executions. Couple the corresponding hidden/report transition rows maximally. The probability of a disagreement by the last time is at most $\eta_0 + \sum \eta_t$. A bounded loss changes in expectation by at most that probability. This argument is uniform over every parameter and every legal controller, including full-history controllers. Infima over controllers and common encoders, and maxima over tasks or parameters, are nonexpansive in uniform error. Apply that observation once to the constrained risk and once to its baseline. A uniformly perturbed loss changes each of these by at most ζ . $\square$

This provides a multi-resource finite theorem with computable synthesis, finite randomization, finite description and finite execution cost. Together with Theorem 5.1, it also makes finite sufficient reconstruction effective whenever the expanded finite source is given rationally. It does not compute multiplicative energies for an arbitrary infinite predictive tree, nor does it turn a merely measurable sufficient statistic into a finite algorithm. Those are different input classes with different resource signatures.

Exact rational reconstruction tables need not have dyadic probabilities. Their computability does not imply exact sampling with a fixed number of fair bits. A fixed-bit implementation rounds their rows and uses Proposition 6.2 to charge the resulting path error; exact non-dyadic sampling requires a different randomization-time convention.

7. A GAUSSIAN EXPERIMENT WITH A GENUINE TEMPORAL PENALTY

The append-register Gaussian theorem in the supporting development concerns a terminal vector experiment. The following experiment instead forces information to survive until the question that uses it arrives. Its final target is scalar, but its causal exponent depends on the number of coordinates acquired before that question.

Fix $d \geq 2$ and $\sigma > 0$. Let $\theta_1, \dots, \theta_d$ be independent and uniform on $[-1, 1]$, and observe sequentially

$$X_j = \theta_j + \sigma Z_j, \quad 1 \leq j \leq d,$$

where the Z_j are independent standard Gaussians. After these observations have been discarded, an independent uniform query $J \in \{1, \dots, d\}$ is revealed. The machine then updates its register and estimates θ_J . There is no external observation transcript. The query is an input to that final update, not an additional persistent decoder register. The cost is squared loss, bounded by four after projecting the answer onto $[-1, 1]$. Write

$$m(x) = \mathbb{E}[\theta_1 \mid X_1 = x], \quad \beta = \mathbb{E} \text{Var}(\theta_1 \mid X_1).$$

Let $\mathcal{E}_S$ be the infimal excess above β when the total register has at most S states at every stage. Independent shared randomness is permitted; the same conclusions hold without it.

Theorem 7.1 (Delayed Gaussian query law). *There are constants $0 < c_{d,\sigma} \leq C_{d,\sigma} < \infty$ such that for every $S \geq 2^d$,*

$$c_{d,\sigma} S^{-2/d} \leq \mathcal{E}_S \leq C_{d,\sigma} S^{-2/d}. \quad (7.1)$$

The same order holds for the maximum over the d individual coordinate query risks, with a common encoder before the query.

In contrast, an encoder which receives the query together with the full observation vector at its sole checkpoint has excess of order S^{-2} . Thus the scalar terminal target has a strictly worse cardinality exponent under the delayed-query causal constraint. The lower bound uses the register before the query and is not a final-output dimension bound.

Proof. The posterior density on $[-1, 1]$ is proportional to $\exp(-(x - u)^2/(2\sigma^2))$. Differentiation under the integral on this compact interval gives

$$m'(x) = \frac{\text{Var}(\theta_1 \mid X_1 = x)}{\sigma^2} > 0. \quad (7.2)$$

The posterior variance is positive because its density is positive on an interval. The function m is smooth, odd, and strictly increasing. Choose $b > 0$ and put $a = m(b) > 0$. The density of X_1 is positive and continuous, and m' is positive and continuous on $[-b, b]$. The change-of-variables formula therefore gives a density for $m(X_1)$ which is bounded below by a positive constant on $[-a, a]$. Independence implies that $M = (m(X_1), \dots, m(X_d))$ has density bounded below by a positive constant on the cube $Q = [-a, a]^d$.

We record an elementary volume bound. If $c_1, \dots, c_S$ are arbitrary points in $\mathbb{R}^d$, and v_d is the volume of the unit ball, choose $r = (|Q|/(2Sv_d))^{1/d}$. The union of their radius- r balls has volume at most $|Q|/2$. Consequently

$$\int_Q \min_{i \leq S} \|x - c_i\|^2 dx \geq \frac{|Q|}{2} \left(\frac{|Q|}{2Sv_d} \right)^{2/d}. \quad (7.3)$$

This bound does not require cells to be convex or centres to lie in Q .

Consider any admissible machine and condition on its independent shared random seed, when present. Its pre-query register I_d has at most S values. Subsequent private updates and readouts can do no better than being given (I_d, J) and using its conditional posterior mean. Let $c_i = (\mathbb{E}[\theta_j \mid I_d = i])_{j=1}^d$, with the seed included in all these conditional expectations. The Markov relation from the full observation to the register, and independence of the random seed, imply $c_i = \mathbb{E}[M \mid I_d = i]$. Orthogonality of conditional expectation shows that the resulting excess for the uniform query is

$$\frac{1}{d} \mathbb{E} \|M - c_{I_d}\|^2 \geq \frac{1}{d} \mathbb{E} \min_{i \leq S} \|M - c_i\|^2.$$

For each fixed seed the law of M is unchanged. Its density lower bound on Q and (7.3) give $c_{d,\sigma} S^{-2/d}$. Averaging the seed preserves this lower bound. The argument permits randomized private encoding: replacing its selected centre by the nearest centre is only a lower bound, not an assumption that the encoding is deterministic. The maximum coordinate risk is at least the uniform average, proving its lower bound as well.

For the upper bound put $L = \lfloor S^{1/d} \rfloor \geq 2$ and partition $[-1, 1]$ into L equal intervals. At observation j compute the interval label of $m(X_j)$ and append it in base L to the stored labels. At that time there are at most $L^j \leq S$ states. After the query, replace the whole label by the interval label of the requested coordinate; the terminal decoder outputs its midpoint. This last update leaves only L possible labels and retains no separate query. Its squared excess in every coordinate is at most $L^{-2} \leq 4S^{-2/d}$. Conditional orthogonality again subtracts the same full-observation baseline β . The construction is deterministic and works with or without a shared seed.

For the noncausal checkpoint comparison, the encoder sees (X, J) and quantizes only $m(X_J)$ into S equal intervals. This has excess at most S^{-2} . The one-dimensional version of the same density lower bound and (7.3) gives the matching lower bound, even with randomized encoding and a shared seed. Its S terminal centres are arbitrary real answers in $[-1, 1]$, so no additional discreteness assumption has entered. This proves all assertions. $\square$

The theorem is a Bayesian slice of the parameterized Gaussian experiment with the specified product prior, not a claim about uniform deficiency over the whole Gaussian family. Its constants depend on d and σ . It controls persistent cardinality and finite acquisition count, not the arithmetic cost of evaluating the real posterior mean. Under a finite numerical signature, an approximation of m requires a separate accuracy and computation bound. The obstruction nevertheless holds for every such implementation, since it was proved for the larger unrestricted-arithmetic class.

The lower bound is the same conditional-variance geometry as Corollary 4.2, now evaluated directly for a continuous history law. In the finite theorem, successor compatibility determines which cells can be executed. Here the late query forces all coordinate readouts to factor through the same pre-query cell, and the volume bound computes its cost. This explains both the lossless append-only regime and a regime where forgetting before a future question changes the exponent, without identifying either one with static terminal dimension.

8. A PRIOR-INDEPENDENT EXPERIMENT BEHIND PREDICTIVE COORDINATES

A future-test quotient relative to one Bayesian preparation need not preserve a parameterized experiment. The additional requirement is *likelihood saturation*. This section gives both a terminal construction and the extra recursion needed for a causal reconstruction. All assertions about measurable coordinates are modulo the stated common null sets.

Theorem 8.1 (A saturated posterior quotient). *Let K be compact metric, let X be standard Borel, and let $E = \{P_\theta : \theta \in K\}$ be continuous in total variation. Choose a dense sequence (θ_j) and weights $w_j > 0$ with $\sum_j w_j = 1$. Put $m = \sum_j w_j P_{\theta_j}$ and define*

$$T(x) = \left(w_j \frac{dP_{\theta_j}}{dm}(x) \right)_{j \geq 1} \in [0, 1]^{\mathbb{N}}. \quad (8.1)$$

There is one parameter-independent kernel R such that

$$R(T_\# P_\theta) = P_\theta \quad (\theta \in K). \quad (8.2)$$

Thus E and TE are statistically equivalent. The completed σ -field generated by T is minimal among sufficient statistics. A different dense sequence and strictly positive weights give the same minimal sufficient σ -field modulo m and the same experiment equivalence class, not necessarily the same coordinates. For the terminal cardinality signature, their marked decision spectra are identical for every budget and every family of bounded terminal parameter-decision marks, whose losses depend on the parameter and the action. An additional latent-target mark requires its own joint-law sufficiency condition. No computability or finite recursive representation of T is asserted.

Proof. If $m(B) = 0$, then $P_{\theta_j}(B) = 0$ for all j ; continuity and density imply $P_\theta(B) = 0$ for every θ . Choose Borel versions of the countably many densities. They give $\sum_j T_j = 1$ m -almost surely; modify them on one common null set if necessary. Disintegrate m given T to obtain the standard Borel conditional kernel $R(t, dx)$. For every j , the density of P_{θ_j} is T_j/w_j , so

$$P_{\theta_j}(B) = \int \frac{T_j}{w_j} R(t, B) T_\# m(dt) = R(T_\# P_{\theta_j})(B).$$

For arbitrary θ , choose $\theta_{j_k} \rightarrow \theta$. Both applying T and applying R contract total variation. Taking limits proves (8.2).

For minimality, let S be sufficient, in the conditional-law sense that one kernel R_S is a conditional law of X given S under every P_θ . Mixing shows that it is also the conditional law under m . The factorization

$$\frac{dP_{\theta_j}}{dm}(x) = \frac{d(S_\# P_{\theta_j})}{d(S_\# m)}(S(x))$$

follows by integrating both sides on arbitrary measurable sets using R_S . Thus every coordinate of T is $\sigma(S)$ -measurable. Conversely the preceding disintegration proves sufficiency of T . Mixtures obtained from different dense sequences are mutually absolutely continuous: each dominates every member of E , hence also the other mixture. Minimality then proves equality of the completed sufficient σ -fields.

Finally, any terminal encoder from TE is an encoder from E by composition with T . Conversely, an S -label encoder Q from E is implemented from TE by QR , with exactly the same label laws

for all parameters. This uses only transient reconstruction randomization and at most S terminal labels. It is a kernel assertion, not an algorithm that stores a sampled infinite object in a finite register. The two sets of attainable terminal risks, and hence their spectra, coincide. $\square$

The theorem makes explicit a classical dominated-sufficiency mechanism [14, 12]. Its role here is to identify the precise terminal slice on which the prior-relative and parameter-uniform notions have the same resource profile. A predictive-test quotient has this property when its generated σ -field contains the likelihood coordinates (8.1). Without that inclusion the conclusion is false, even for a two-point parameter set: observe $Y = \theta \in \{0, 1\}$ once, and allow only constant future tests. Their quotient is a point although the observed experiment distinguishes both parameters exactly. Predictive equivalence always refers to a declared family of future questions.

8.1. The causal factorization. We give a finite-horizon formulation to avoid any assertion that a terminal sufficient statistic is automatically a causal sufficient state. Let $H_t = (A_0, Y_1, \dots, A_{t-1}, Y_t)$ be the observed history. All policies have the same parameter-independent action kernels. Under a reference prior on θ , write $\bar{P}$ for the mixture law. Suppose Borel coordinates $S_t = S_t(H_t)$ satisfy the following *saturation conditions*, with versions fixed for every admissible history and action where they are used:

$$S_t = F_t(S_{t-1}, A_{t-1}, Y_t), \quad (8.3)$$

$$\bar{P}(dy \mid H_{t-1}, a) = k_t(dy \mid S_{t-1}, a), \quad (8.4)$$

$$\frac{dP_\theta|_{H_t}}{d\bar{P}|_{H_t}} = \ell_{\theta,t}(S_t). \quad (8.5)$$

The likelihood identity is required for each parameter, initially on a dense reference family and then for all parameters by finite-horizon variation continuity. These conditions hold uniformly over the policies under consideration. Alternatively one may assume domination and the identity directly for every parameter. The reference prior is not allowed to change with the policy during a reconstruction.

Theorem 8.2 (Causal reconstruction of a saturated quotient). *Under (8.3)–(8.5), there are parameter-independent kernels $L_t(dy \mid s, a, s')$ which reconstruct the observation stream from its successive quotient states, with the correct joint law under every admissible policy. Hence the original and quotient instruments have mutually exact causal simulations. If S_t has at most K_0 values at each time, both simulations may be implemented with K_0 internal states and the cardinality spectra obey the budget dilations of Theorem 2.2. If the quotient is not finite, the theorem asserts statistical and causal reconstruction only, with the storage of S_{t-1} explicitly charged; it does not assert a finite-register resource equivalence.*

Proof. Let $H_t^0(ds' \mid s, a)$ be the pushforward of $k_t(\cdot \mid s, a)$ by $F_t(s, a, \cdot)$. Disintegration, including the Borel variables (s, a) , gives

$$k_t(dy \mid s, a) = \int L_t(dy \mid s, a, s') H_t^0(ds' \mid s, a), \quad (8.6)$$

with $F_t(s, a, y) = s'$ under the resulting joint kernel. On histories with $\ell_{\theta,t-1}(s) > 0$, change of measure gives the actual joint conditional law of (Y_t, S_t) as

$$L_t(dy \mid s, a, s') \frac{\ell_{\theta,t}(s')}{\ell_{\theta,t-1}(s)} H_t^0(ds' \mid s, a).$$

In particular, conditioning further on s' removes the likelihood factor. On zero-likelihood histories the kernel may be chosen arbitrarily; such histories are not visited under that parameter. For parameters outside a countable reference support, equality of the constructed finite-horizon laws extends by total variation, or follows directly from the assumed domination identity.

Start at the declared initial state. Given a target policy on the reconstructed reports, feed its chosen action into the quotient instrument; on receiving s' , draw y with $L_t(\cdot \mid s, a, s')$, then replace s by s' . Induction proves equality of every finite parameter-indexed report/action law. This induction is essential: it reconstructs responses after the same actions, not after actions selected with hidden parameter information. In the forward direction, apply F_t recursively to the real reports. Both executions retain only the previous quotient state in addition to the target encoder's own register. This costs $K_0 S$ states if that state has K_0 possible values. The general standard Borel version has no such finite bound, exactly as stated. $\square$

A likelihood statistic alone supplies (8.5), not (8.4). Conversely, a Bayesian predictive state supplies future report predictions, not necessarily all parameter likelihoods. The conjunction is the bridge. Sequential comparison of filtered experiments already has a substantial decision-theoretic history [15]; the present statement fixes the additional register cost of the reconstruction rather than claiming that sequential comparison itself is new.

For marks involving an additional latent target rather than the parameter, its conditional kernel must also be included in the predictive saturation condition. The causal reconstruction theorem does not infer equality of latent-target laws from equality of report laws alone.

9. A NECESSITY THEOREM FOR CAUSAL REALIZATION

We now characterize the causal constraint without imposing a tree geometry. The time dependence of a machine is retained, so the result does not replace a free clock by an uncharged stationary-state assumption. Fix a horizon T , finite report alphabets, one Bayesian law, and square-integrable Hilbert targets U_t . There is no control in this section. Let $\mathcal{H}_t$ be the finite set of positive-probability histories, $p_t(h) = \mathbb{P}(H_t = h)$, and $m_t(h) = \mathbb{E}[U_t \mid H_t = h]$. Let $\beta_t \geq 0$ be fixed weights.

A family of partitions $(\mathcal{P}_t)_{t=0}^T$, with $\mathcal{P}_0 = \{\emptyset\}$, is *successor compatible* if

$$h, h' \in C \in \mathcal{P}_t, \quad hy, h'y \in \mathcal{H}_{t+1} \implies hy, h'y \text{ belong to the same cell of } \mathcal{P}_{t+1}. \quad (9.1)$$

Only common positive-probability continuations enter this condition. For $C \in \mathcal{P}_t$ put $\bar{m}_C = p_t(C)^{-1} \sum_{h \in C} p_t(h) m_t(h)$ and define

$$\mathcal{V}(\mathcal{P}) = \sum_{t=1}^T \beta_t \sum_{C \in \mathcal{P}_t} \sum_{h \in C} p_t(h) \|m_t(h) - \bar{m}_C\|^2. \quad (9.2)$$

Theorem 9.1 (Recursive partition characterization). *For the cardinality signature, allowing randomized time-dependent updates and decoders and independent public randomness, the optimal weighted excess over the full-history Bayes risk equals*

$$\inf_{\#I_t \leq S} \sum_{t=1}^T \beta_t \mathbb{E} \|m_t(H_t) - \hat{U}_t\|^2 = \min_{\substack{\mathcal{P} \text{ successor compatible} \\ |\mathcal{P}_t| \leq S}} \mathcal{V}(\mathcal{P}). \quad (9.3)$$

Thus same-order realization of a sequence of static checkpoint optima is equivalent to the existence of successor-compatible partitions with the corresponding widths and variance bounds. This is a necessary as well as a sufficient condition. It is invariant under relabelling of histories; suffix closure is one sufficient construction of these partitions, not a necessary choice of coordinates.

Proof. A deterministic initialized machine partitions $\mathcal{H}_t$ according to I_t . Its update depends only on (t, I_t, y) , so its partitions satisfy (9.1). Given a state cell, the mean $\bar{m}_C$ minimizes its squared error by Hilbert orthogonality. The resulting excess is precisely (9.2).

Conversely, label each partition's cells by at most S labels. For each cell C and symbol y , send its label to the cell containing hy for any supported $h \in C$ with supported extension hy .

Compatibility makes the answer independent of the choice of h . If there is no such extension, define the update arbitrarily. At time t decode the label as $\bar{m}_C$. Induction identifies the actual label with the cell of the actual history. There are finitely many partition families, so the minimum exists.

To include randomization, realize the finitely many update and decoder kernels by independent uniforms, and condition on their entire tape. For each tape, the rule is a deterministic time-dependent S -state machine. Independence of the tape and the experiment preserves the history law and conditional means. The average of its weighted costs cannot be smaller than the deterministic minimum. Public independent randomness is handled in the same way. This freezing argument concerns one fixed Bayesian sum; it does not interchange a minimax task supremum with the average over tapes.

Unrestricted checkpoint encoders correspond to arbitrary partitions at each time, without (9.1). Hence they have cost $\sum_t \beta_t e_S^2(m_t(H_t))$. Formula (9.3) gives exactly the asserted equivalence of order comparisons, including the case when a static cost is zero. $\square$

Corollary 9.2 (Finite-command cell instruments). *Let $k_1, \dots, k_J$ be Borel functions on a parameter space with $k_j \geq 0$ and $\sum_j k_j = 1$, and fix a prior μ . At each time the raw cell is drawn independently conditional on the parameter. A command g from a prescribed finite menu in $[\eta, 1-\eta]^J$, $0 < \eta < 1/2$, has a prescribed parameter-independent law ν_t . Cell j is reported as j with probability g_j and otherwise as the single failure symbol $\dagger$. Only the command and the reported symbol are observed. Fix a finite horizon, finite future-query menus with known likelihoods $H_{tq} \in [0, 1]$, independent query weights ω_{tq} , and nonnegative checkpoint weights β_t .*

For the persistent-cardinality signature, the optimal common causal encoder for the weighted sum of quadratic prediction excesses is exactly (9.3), with the finite history law and query means computed from the actual cell likelihoods. This applies to the finite-command specialization of the A1 finite-cell protocol [19]. It is an exact scalarized criterion, not an asserted solution of the different maximum-over-checkpoints criterion with private randomization.

Proof. For the observed pair (g, x) define

$$\lambda_{g,x}(\theta) = \begin{cases} g_x k_x(\theta), & x \in \{1, \dots, J\}, \\ \sum_j (1 - g_j) k_j(\theta), & x = \dagger. \end{cases}$$

For a history $h = ((g_s, x_s))_{s \leq t}$ its probability is $\prod_{s \leq t} \nu_s(g_s)$ times $\int \prod_{s \leq t} \lambda_{g_s, x_s}(\theta) \mu(d\theta)$. On positive-probability histories the mean of query q is

$$m_{tq}(h) = \frac{\int H_{tq}(\theta) \prod_{s \leq t} \lambda_{g_s, x_s}(\theta) \mu(d\theta)}{\int \prod_{s \leq t} \lambda_{g_s, x_s}(\theta) \mu(d\theta)}.$$

Take the Hilbert prediction vector with coordinates $\sqrt{\omega_{tq}} m_{tq}(h)$ and apply Theorem 9.1. All histories are finite, and the law used there is exactly the actual acquisition law above. A failure history contains $\dagger$, not the unobserved cell j ; the successor compatibility condition therefore requires one common update for that report. Integration of the given Borel likelihoods may require arbitrary real constants or offline work. Neither is claimed finite under the cardinality signature. Zero-evidence histories are omitted rather than given artificial posteriors. $\square$

Corollary 9.3 (Continuation obstruction). *Let g be a bounded readout on all finite words over a finite alphabet. For stationary deterministic machines required to reproduce g on every word, an exact S -state realization exists if and only if the relation*

$$h \equiv_g h' \iff g(hw) = g(h'w) \text{ for every finite } w \quad (9.4)$$

has at most S classes. More generally put $d_(h, h') = \sup_w \|g(hw) - g(h'w)\|$. Every stationary deterministic realization with uniform error at most ϵ has state fibres of d_* -diameter at most 2ϵ , and thus at least as many states as any 2ϵ -separated set of histories.*

Proof. Histories in one machine state have the same state after every common continuation. Exact output equality implies (9.4); for approximate outputs the triangle inequality gives the diameter bound. The relation (9.4) is a right congruence. Its classes, updated by appending the new symbol, form an exact machine, whose output is $g(h)$, well defined by the empty continuation. This is the output version of the classical finite-automaton congruence argument [16]; the finite-horizon, weighted-distortion characterization in Theorem 9.1 is the form needed here. $\square$

Example 9.4 (A two-valued checkpoint requiring exponentially many persistent states). At the first time observe a uniformly distributed word $X \in \{0, 1\}^d$. At the second time observe an independent uniform index $J \in \{1, \dots, d\}$ and predict $U = X_J$. The raw first acquisition costs d bits; it is not called a one-bit observation. The full-history conditional mean at the second checkpoint has only two values, so its static two-label error is zero. Nevertheless a causal exact predictor requires $S \geq 2^d$. Two different first-stage words in one state differ in a coordinate which can be queried with positive probability. The second-stage update cannot separate them. Retaining the whole word is sufficient.

There is also a quantitative obstruction. If e is the smallest error probability of an S -state predictor, $2 \leq S \leq 2^d$, then

$$e \geq h_2^{-1} \left(1 - \frac{\log_2 S}{d} \right), \quad (9.5)$$

where h_2 is the binary entropy and its inverse takes values in $[0, 1/2]$. Indeed, condition also on every independent public coding variable R . Since the first register has at most S values, $H(X \mid I_1, R) \geq d - \log_2 S$. Predicting each coordinate optimally from (I_1, R) gives errors $e_j \leq 1/2$, and the binary entropy inequality and concavity give

$$H(X \mid I_1, R) \leq \sum_{j=1}^d H(X_j \mid I_1, R) \leq \sum_{j=1}^d h_2(e_j) \leq d h_2 \left(d^{-1} \sum_j e_j \right).$$

The average on the right is a lower bound for any actual queried-bit error. This proves (9.5). If squared loss is used instead, its optimal value is at least $e/2$, since $p(1-p) \geq \min(p, 1-p)/2$. Thus the discrepancy persists for the Hilbert-loss slice of the spectrum. It is a temporal obstruction, not poor scalar quantization. The query must be answered after the update; J is not an additional persistent decoder input.

10. FINITE-PREFIX, FINITE-DESCRIPTION REALIZATION

The recursive partition theorem isolates a necessity condition. Predictive-energy trees provide a verifiable sufficient construction. The full tree theorem and its geometric hypotheses are proved in Section 15 below. We now strengthen its implementation: exact ordering of nearly tied real energies is unnecessary, and the raw acquisition need not expose an infinite word.

Use the factorial legal language, stationary cylinder masses p_v , energies a_v , scales $D_v = (a_v/p_v)^{1/2}$, and constants a_-, ρ, U, C_0, c_0 of that theorem. In particular $a_\emptyset = 1$, $a_{vi} \leq \rho a_v$ with $\rho < 1$, $a_{vi} \geq a_- a_v$, $D_{vi} \leq D_v$, and $a_{uv} < a_v$ for nonempty u . Legality and the child list are decidable. Suppose an oracle returns a positive rational upper estimate u_v satisfying, for a fixed $\kappa \geq 1$,

$$a_v \leq u_v \leq \kappa a_v. \quad (10.1)$$

No uniform cost for this oracle is hidden in a cardinality assertion. Set

$$\bar{a}_v = \min\{u_w : w \text{ is a contiguous factor of } v, \text{ including } \emptyset\}. \quad (10.2)$$

For $0 < \eta < 1$, split precisely the legal vertices for which $\bar{a}_v > \eta$. Retain all their legal children. Denote the resulting full finite tree by $\mathcal{T}_\eta$, its leaf cut by $\mathcal{C}_\eta$, and its number of leaves by l_η .

Theorem 10.1 (Robust tree realization with charged acquisition). *The rational-threshold construction is finite, suffix closed, and supports both deletion and prepend-and-parse insertion. Its depth and state count satisfy*

$$D_\eta \leq 1 + \left\lceil \frac{\log(\kappa/\eta)}{|\log \rho|} \right\rceil, \quad (10.3)$$

$$V_\eta = 1 + \sum_{v \text{ internal}} \deg^+(v) \leq (U+1)(2l_\eta - 1). \quad (10.4)$$

For constants depending only on the displayed structural bounds and κ ,

$$\sum_{v \in \mathcal{C}_\eta} a_v \asymp e_{l_\eta}^2(Z) \asymp G_{l_\eta}. \quad (10.5)$$

In the deletion experiment, at most D_η symbols of a snapshot acquisition suffice; no later access to that snapshot is used. The same average-risk order as the exact greedy construction follows. For every sufficiently large state budget S , a dyadic threshold can be selected with $V_\eta \leq S$, $l_\eta \geq cS$, and $D_\eta = O(\log(S+1))$.

If the online readouts are in a fixed bounded subset of $\mathbb{R}^d$ and representatives can be computed to prescribed accuracy, the selected machine has a finite rational presentation. Readout precision $p = O(\log(S+1))$ suffices to retain the risk order. A word list, transition table and readout table have total description

$$P = O(S \log(S+1) + Sdp) \quad (10.6)$$

for a fixed alphabet. Its online persistent storage is one of at most S states; a transition-table implementation uses transient storage $O(\log(S+1) + dp)$ bits. An acquisition uses $O(\log(S+1))$ symbol queries. Online table accesses can be implemented in word-RAM time proportional to these query counts; a sequential bit-table implementation gives the conservative bound $O(S(\log(S+1) + dp) \log(S+1))$ bit operations per acquisition. Offline oracle time is separate. The Hilbert-space cardinality assertion does not, by itself, encode an infinite-dimensional readout in finitely many bits.

Proof. If w is a factor of v , removing a prefix and then a suffix cannot decrease energy. Consequently

$$a_v \leq \bar{a}_v \leq \kappa a_v, \quad \bar{a}_{iv}, \bar{a}_{vi} \leq \bar{a}_v. \quad (10.7)$$

The second assertion follows by inclusion of factor sets and requires no accurate comparison between two distinct true energies. Thus the internal vertices form a prefix-closed and suffix-closed set. Each internal vertex is split into all its legal children. A proper suffix of a leaf is a legal child of a suffix of its parent, hence is present; repeating this argument proves full suffix closure. If iv had no leaf prefix when v is a leaf, it would be internal, forcing its suffix v to be internal, a contradiction. These are exactly the two update operations required.

At depth n , $\bar{a}_v \leq \kappa \rho^n$, which proves finiteness and (10.3). Every nonroot leaf has an internal parent, so

$$a_{-}\eta/\kappa < a_v \leq \eta. \quad (10.8)$$

The exact vertex count follows by counting edges; contracting unary chains gives (10.4) as in the full-tree lemma below.

Here is a lower bound not depending on exact greedy ordering. Refine all leaves by $h = 3(U+1)$ further legal levels. Each leaf has at least 2^3 descendants at that level, because there are at most U consecutive unary vertices. The resulting incomparable cylinders number at least $8l_\eta$. Every one has energy at least $a_{-}^{h+1}\eta/\kappa$. The separation hypothesis and monotonicity of D make the open tubes of radii δD_w about their images disjoint for $2\delta < c_0$. At most l_η tubes can contain one of l_η arbitrary proposed Hilbert centres. Each unoccupied cylinder contributes at least $\delta^2 a_w$ to the squared error. Hence $e_{l_\eta}^2(Z) \geq c_l \eta$. One representative per original leaf gives

$e_{l_\eta}^2(Z) \leq C_0^2 \sum a_v$, and (10.8) proves the first comparison in (10.5). The second is the static conclusion of Theorem 15.1, whose proof does not use the present theorem. Its oscillation estimate in the deletion orientation, or its cylinder diameter estimate in the insertion orientation, also proves the stated risk bounds for this cut.

A prefix parser stops on reaching a leaf, after at most D_η queries. It never reads or stores the rest of the snapshot. This is a finite-latency acquisition convention: the snapshot is available during that update and the prediction deadline permits these microsteps. No claim is made for a moving physical state observed after an unmodelled read delay. Insertion and deletion then use the already proved closure.

For budget selection, use $\eta_j = 2^{-j}$ and take the last threshold with $V_{\eta_j} \leq S$. There is a constant bound on $V_{\eta_{j+1}}/V_{\eta_j}$. In fact, beneath a leaf at threshold η , all descendants at depth k have estimated energy at most $\kappa\rho^k\eta$, and so the next threshold adds at most k levels once $\kappa\rho^k \leq 1/2$. The ratio is bounded by a fixed geometric sum of powers of the alphabet size. It follows that the chosen V_η is at least a fixed multiple of S , and then (10.4) gives $l_\eta \geq cS$. For completeness, a leaf at depth n has $a_v \geq a_-^n$, so $n \geq \log(1/\eta)/|\log a_-|$. Every legal path branches at least once in $U+1$ levels. The complete cut therefore has at least $2^{\lfloor \log(1/\eta)/((U+1)|\log a_-|) \rfloor}$ leaves. Since $l_\eta \leq S$, this bounds $\log(1/\eta)$ by $O(\log(S+1))$. Combine with (10.3). Fixed-dilation comparison for G then converts (10.5) into the same- S order.

Finally replace each finite-dimensional readout by a rational vector within Euclidean distance δ . Pointwise squared loss is at most twice its previous value plus $2\delta^2$. Take $\delta^2 \leq \eta$ and use $\sum a_v \geq a_- l_\eta \eta / \kappa$. Bounded readouts require $p = O(\log(1/\eta) + \log(d+1))$ binary digits per coordinate. Store all vertex words, their legal child/deletion/insertion transitions, and these rational vectors. There are $O(S)$ transitions for a fixed alphabet, each state index costs $O(\log(S+1))$ bits, and the word-list depth is $O(\log(S+1))$. This proves (10.6). A streaming parser uses only its current vertex index during the acquisition; no prefix string survives the update. The stated time bounds follow by random access, or by a full sequential scan for each table access. The number and precision of the offline queries can be listed, but (10.1) alone gives no uniform running-time bound for answering them. This is why offline time is not absorbed into (10.6). $\square$

The use of all factors in (10.2) is essential. Independent relative approximations u_v need not respect suffix monotonicity, even when the true energies do. Thresholding the raw approximations can therefore produce a tree that cannot execute its own suffix update. The envelope repairs that defect while changing energies by at most the fixed factor κ .

11. THE MEASURE-VALUED SLICE: SHARP PRESENTATION ENTROPY

The decoder's output type is part of a mark. In particular, outputting a probability law close in expected Wasserstein distance is stronger than simulating one randomly chosen observation from that law. This separation is important for a genuinely measure-valued predictive state.

Let $\mathcal{M} = \mathcal{P}([0,1])$ with W_1 . A *uniform law presentation* receives $\pi \in \mathcal{M}$ as a transient input and returns one of S persistent labels. Its decoder outputs a law $\hat{\pi}$. The fixed code, including its representative laws, is chosen once for all π . In particular the unknown input is not inserted into the decoder's free program constants. Let

$$H_S = \inf_{Q,D} \sup_{\pi \in \mathcal{M}} \mathbb{E} W_1(\pi, D(Q(\pi))). \quad (11.1)$$

Both maps may use independent private or public randomness. The criterion is the expected distance of the *realized* decoded law. Only persistent cardinality is charged here; the input-law interface and general real-law computation are unrestricted. Regard π as the unknown parameter and the uncompressed observation as π itself. Then H_S is the spectrum of the single bounded minimax law-presentation mark, with full-observation baseline zero. It is not a supremum of

point-prior Bayes risks: such a prior would reveal the unknown law to the decoder and change the problem.

Theorem 11.1 (Uniform presentation of a probability measure). *There are absolute $c, C > 0$ such that*

$$\frac{c}{\log(S+1)} \leq H_S \leq \frac{C}{\log(S+1)}, \quad S \geq 1. \quad (11.2)$$

Moreover no continuous map from $\mathcal{M}$ into a fixed $\mathbb{R}^k$ can be an exact coordinate sufficient to evaluate every bounded Lipschitz integral on all of $\mathcal{M}$. This statement does not exclude noncontinuous Borel encodings of a standard Borel space into one real coordinate.

Proof. For the upper bound use the explicit net in Lemma 17.1 below: equally weighted n -point empirical laws with points on the grid $\{0, 1/n, \dots, 1\}$, in increasing order. There are $\binom{2n}{n} \leq 4^n$ such laws. Quantile discretization followed by grid rounding gives covering radius $3/(2n)$. Taking $n = \lfloor \log S / \log 4 \rfloor$ proves the upper bound for $S \geq 4$; increase C for smaller S .

For the lower bound, partition $[0, 1]$ into n equal intervals. In each interval put mass $1/n$ at either its quarter point or its three-quarter point. Denote the resulting law by π_ω , $\omega \in \{0, 1\}^n$. Their quantile functions give the exact identity

$$W_1(\pi_\omega, \pi_{\omega'}) = \frac{d_H(\omega, \omega')}{2n^2}. \quad (11.3)$$

Greedily choose binary words and delete Hamming balls of radius $\lfloor n/4 \rfloor$. The selected words have mutual distance greater than $n/4$, and their number is at least $2^{c_0 n}$, where $c_0 = 1 - h_2(1/4) > 0$. Indeed $\sum_{j \leq n/4} \binom{n}{j} \leq 2^{nh_2(1/4)}$, as follows by bounding these binomial terms using the binomial expansion at parameter $1/4$. Thus the selected measures have mutual W_1 distance greater than $1/(8n)$.

Choose $n \leq C_0 \log(S+1)$ sufficiently large that the packing has at least $2S$ members. Give it the uniform prior. For any S proposed law centres, their open balls of radius $1/(16n)$ contain at most S packing members in total. At least half the prior therefore incurs distance at least $1/(16n)$, no matter how it is encoded. Fixing all independent coding randomness gives this same bound for each tape; integration preserves it. The worst input risk is at least the uniform prior risk, which proves the lower bound in (11.2).

For the last assertion, bounded Lipschitz integrals separate probability measures, so a sufficient coordinate map must be injective. The measures supported on any $k+2$ distinct points form a $(k+1)$ -simplex. A small sphere about its barycentre lies in its relative interior and is homeomorphic to S^k . By the Borsuk–Ulam theorem every continuous map of this sphere to $\mathbb{R}^k$ identifies two antipodal points [18]. Hence no such injective coordinate exists. For $k=0$ two distinct measures already give the contradiction. $\square$

The entropy mechanism is consistent with the power-exponential geometry of Wasserstein spaces studied in [17]; the packing argument is supplied here because the exact presentation loss, including randomization, matters for the result.

Corollary 11.2 (A nonlinear posterior class with a matching bound). *Consider the nonlinear binary instrument of Corollary 17.3 below, with*

$$h_0(x) = x/5 + x^2/50, \quad h_1(x) = 3/5 + h_0(x), \quad \Phi_y \pi = \frac{3}{4}(h_y)_\# \pi + \frac{1}{4}(h_{1-y})_\# \pi.$$

For each fixed report y ,

$$\frac{1}{5} W_1(\pi, \nu) \leq W_1(\Phi_y \pi, \Phi_y \nu) \leq \frac{6}{25} W_1(\pi, \nu). \quad (11.4)$$

Consequently, for every fixed t , a single finite-register scheme that must present the posterior for every input prior has worst expected W_1 error at least $c5^{-t}/\log(S+1)$ at time t , even if the decoder

is additionally told the reports. A uniform recursive presentation has error $O(1/\log(S+1))$ at all times, when the initial law is encoded at the same accuracy and the update error is of that order. No fixed finite-dimensional continuous exact predictive coordinate exists on this all-priors posterior class. Constants in the lower bound are not asserted uniform as $t \rightarrow \infty$ for one fixed preparation.

Proof. The images of h_0 and h_1 lie in two disjoint ordered intervals. The mixture weights for a fixed y agree for π and ν . Quantile coupling therefore gives

$$W_1(\Phi_y \pi, \Phi_y \nu) = \frac{3}{4} W_1((h_y)_\# \pi, (h_y)_\# \nu) + \frac{1}{4} W_1((h_{1-y})_\# \pi, (h_{1-y})_\# \nu).$$

The derivatives of both increasing maps lie between $1/5$ and $6/25$, proving (11.4). Iterate it along any length- t report word and apply the same packing as in the preceding proof. The reports of this instrument are independent fair bits, with a law independent of the initial prior. Conditioning on the report word thus does not change the uniform packing prior. The centre-counting lower bound survives even with this word supplied freely to the decoder; integrating it over words and coding tapes proves the assertion for the actual, less informed decoder.

The recursive upper bound follows from Theorem 17.2 with $\rho = 6/25$, initializing by a net point rather than by an arbitrary fixed law. The initial error is at most $3/(2n)$ and the accumulated error is bounded by $[3/(2n) + \eta]/(1 - 6/25)$. Every fixed-word map on laws is continuous and injective by the lower bound in (11.4). Its restriction to the compact simplex used above is a homeomorphism onto its image. Applying the same sphere argument proves the coordinate obstruction. $\square$

Proposition 11.3 (Marginal-probe simulation is a different presentation problem). *Suppose only one Bernoulli probe is required: after compression a function $g : [0, 1] \rightarrow [0, 1]$ with Lipschitz constant at most one is specified, and the desired output law is $\text{Bern}(\pi g)$. The probe is fixed independently of the realized encoder randomness. For $S \geq 2$ there is a common S -label encoder whose output-law error is at most $1/[2(S-1)]$, uniformly in π, g . Thus this marginal simulation mark cannot be substituted for the realized-law mark in (11.1).*

Proof. Sample $X \sim \pi$ transiently, store the nearest point Z of the S -point uniform grid, and discard X . On receiving g , output a Bernoulli variable with parameter $g(Z)$. Its unconditional law is Bernoulli with parameter $\mathbb{E}g(Z)$, and $|\mathbb{E}g(Z) - \pi g| \leq \mathbb{E}|Z - X| \leq 1/[2(S-1)]$. The encoder is chosen before g . It does not produce a realized law close to π in W_1 : its natural decoded law is δ_Z . Nor does the argument cover probes chosen after observing Z , or an unlimited transcript of independent samples. Those are different tasks and interfaces. This output-law simulation criterion is not identified with a realized-law decision loss. $\square$

12. A DYNAMIC FINITE-EXPERIMENT CONSEQUENCE

The decision-complete slice of the same spectrum has a different geometry. We first make its temporal register convention explicit, and then connect it to the physical count experiment.

Fix $n, r \geq 1$, a compact $K \subset \mathbb{R}^r$ with nonempty interior, and a positive definite covariance Γ . At successive times observe independent

$$Y_t \sim N(\theta_t, \Gamma), \quad \theta = (\theta_1, \dots, \theta_n) \in K^n. \quad (12.1)$$

Each Y_t is discarded after the update. No earlier output labels are retained outside the register. At the terminal time only its final label is supplied to a parameter-independent reconstruction kernel. This is not an S -symbol channel whose entire transcript is available.

Theorem 12.1 (Sequential accumulation under a total register budget). *For the cardinality signature with persistent S states and at most S terminal outcomes, the decision-complete terminal spectrum of (12.1) satisfies*

$$c_n S^{-2/(nr)} \leq \Theta_{G(n)}(S; \mathcal{D}_{\text{all}}) \leq C_n S^{-2/(nr)}. \quad (12.2)$$

The infimum is over causal encoders obeying the total state budget at every time. Lower bounds include arbitrary randomized, time-dependent encoders. A deterministic causal encoder attains the upper order. Constants are for fixed n, r, K, Γ , not uniform in a growing horizon. The readout/reconstruction randomness is external and fresh and private, while the physical alphabet and the persistent register remain finite. No public tape or other side information is supplied alongside the final label in this theorem.

Proof. The full observation vector is a Gaussian location experiment of dimension nr , with compact mean set K^n and covariance $I_n \otimes \Gamma$. Every final register gives an experiment with at most S outcomes, regardless of its causal implementation. The lower bound of Theorem 18.1, proved below, applies to all such experiments. Because a register is a garbling of the full observation, its forward deficiency is zero. Theorem 2.4 therefore identifies its reverse deficiency with its worst normalized Bayes regret, proving the lower bound in (12.2).

For the upper bound put $J = \lfloor S^{1/(nr)} \rfloor$. At time t compactify each coordinate of Y_t by $\tau(x) = 1/2 + \arctan(x)/\pi$, quantize into J cells per coordinate, and append the resulting r -tuple of cell indices to the current register word. At that time there are $J^{rt} \leq J^{rn} \leq S$ possible words. The known absolute time determines their length; it carries no data-dependent information. Thus a single S -value register, not n separately uncharged registers, implements the rule. At the final time apply the positive tensor-hat reconstruction of Lemma 18.3 in dimension nr . For $J \geq 2$ its error is $O(J^{-2}) = O(S^{-2/(nr)})$. Enlarge the constant for the remaining finite budget range. This proves the upper bound using a genuine causal encoder. Its real-valued input computation has no finite-time claim under this signature. $\square$

If instead the sequence of emitted cell labels were supplied to the terminal decoder, its alphabet would be their product. In that different model a small current register could leave a large data-bearing transcript. The lower bound above is about the total retained experiment, not the size of the most recent communication symbol. This is one reason zero-delay channel coding and total-register coding require separate resource signatures [3, 4].

12.1. A theorem-level downstream arrow. We now give the adapter in the coordinates of the retained known-mark exposure protocol, rather than identifying all of the algebraic paper with this statistical subexperiment.

Corollary 12.2 (Known-mark exposure adapter). *Fix positive exposure centre λ^0 , known marks, a derivative matrix D of full row rank, and $\Lambda = \text{diag}(\lambda^0)$. Put $\Sigma = D\Lambda D^\top$ and $B_* = \Lambda D^\top \Sigma^{-1}$. Let L be a fixed normal-compression map of rank $r \geq 1$, let F be an orthonormal frame for its range, and set*

$$C = F^\top L D, \quad A = F^\top L, \quad \Gamma = C \Lambda C^\top.$$

*On a fixed compact index set H , observe independent counts of means $N\lambda_N(h)$, where $\lambda_N(h) = \lambda^0 + N^{-1/2}B_*h$. If AH contains an r -dimensional ball, the deterministic unjittered statistic $T_{N,J}$ of Section 19, applied to the score $C(N - N\lambda^0)/\sqrt{N}$, has at most $J^r \leq S$ outputs. For $\mathcal{G} = \{N(Ah, \Gamma) : h \in H\}$, it satisfies*

$$cS^{-2/r} \leq \Delta(\mathcal{E}_{N,J}, \mathcal{G}) \leq C_K S^{-2/r}, \quad J = \lfloor S^{1/r} \rfloor, \quad 2^r \leq S \leq KN^{r/6}. \quad (12.3)$$

Only the physical output cardinality and the stated joint sample/register window are bounded here. Reconstruction randomization is external, parameter independent and private; no finite-bit complexity of the raw real score computation is asserted.

Proof. $DB_* = I$, so $CB_* = A$. Since $F^\top L$ has row rank r and D is surjective, C has row rank r and $\Gamma > 0$. The compact local alternatives have uniformly positive exposures for all sufficiently large N . Thus every hypothesis of Theorem 19.1 is verified with these actual score weights, mean set and covariance. That theorem supplies both inequalities in (12.3); its lower bound

allows every experiment with at most S outcomes, not only this statistic. Theorem 2.4 gives the all-decision interpretation of the limiting obstruction, and Theorem 2.2 transports each bounded task through the comparison kernels. No algebraic primary-decomposition theorem is a premise of this argument. $\square$

The left inequality concerns every experiment with at most S terminal outputs approximating this specified Gaussian family, and therefore also the physical statistic. The upper inequality concerns that explicit statistic and its positive reconstruction kernel. In particular, achieving Gaussian-family distance at most $CN^{-1/5}$ forces $S \geq cN^{r/10}$, and budgets of that order suffice for this statistic. These are label-budget statements in the displayed joint window; they are not a sample-optimality theorem over all finite- N physical objectives.

Theorem 2.4 supplies a decision interpretation of the limiting experiment's finite-output obstruction, while Theorem 2.2 transports each bounded task through the Poisson comparison kernels. These are genuine arrows between specified experiments. They neither require nor assert a Sinai local limit theorem, a particle path large-deviation principle, or a physical phase construction.

SUPPORTING THEORY AND MODEL VERIFICATION

The following sections give the complete proofs used above. Unless a more detailed signature is specified, they use the cardinality signature of Section 2: only persistent states or physical terminal outcomes are bounded, with arbitrary transient computation and independent reconstruction randomness. In the tree application, Theorem 10.1 additionally replaces the full-word acquisition by a charged finite-prefix interface. In the measure application the new matching lower bound is for the uniform realized-law presentation task, not for a single marginal probe. The Gaussian and Poisson inequalities concern the indicated terminal experiment family. These local conventions are part of their use here.

13. POSITIVE EXPERIMENTS AND PREDICTIVE QUOTIENTS

13.1. Actual histories. All spaces of states, observations and actions are standard Borel. For an unknown parameter λ , a controlled experiment has a probability kernel

$$K_t^\lambda(dx', dy, dc \mid x, a), \quad (13.1)$$

where c is the cost of the actual step. Failure, rejection and stopping are outcomes, not missing probability mass. A policy chooses a_t from the observed past and independent randomization. Preparation has a specified law and cost. With an absorbing symbol after stopping, repeated kernel integration constructs one probability law on the infinite history space. Its finite marginals are consistent because every kernel in (13.1) has mass one. This is the usual kernel extension theorem [2]; conditioning on success would define a different experiment unless its preparation and waiting costs were separately charged.

For a fixed Bayesian preparation, let π be a reachable belief state on the hidden state, including the parameter when it is random. The positive instrument associated with an action and observation event is

$$\mathsf{T}_{a,B}\pi(E) = \int K(E, B, dc \mid x, a) \pi(dx).$$

After an observed event of positive probability, normalization gives the updated belief. For nonatomic reports we use a specified disintegration, with all equalities interpreted outside the appropriate null sets. A formula for an event of probability zero is not an additional observation channel.

13.2. The future-test quotient. A future test consists of a finite observation-based policy, a stopping rule, and a bounded measurable terminal payoff. Its conditional expectation at a belief π is written $\pi(v)$, where v is the corresponding backward test function. Fix a family $\mathcal{V}$ containing constants and closed under the one-step pullbacks of the allowed tests. It may, for example, contain all bounded finite-horizon executable tests. Define

$$\pi \sim_{\mathcal{V}} \pi' \iff \pi(v) = \pi'(v) \quad (v \in \mathcal{V}). \quad (13.2)$$

The set of classes is the predictive quotient. The choice of tests is part of the experiment specification: identifying a hidden state is not required if none of the designated tests distinguishes it.

Proposition 13.1 (Minimality and descent). *The quotient (13.2) is the coarsest set-valued statistic that determines all designated predictions. If the test family is one-step closed, normalized Bayesian updates descend to the quotient where their normalization is positive. If, in addition, the reachable belief domain B is compact metrizable, a countable family of continuous tests determines (13.2), and each fixed-report update is continuous on its positive-probability domain, the quotient has a canonical metrizable realization as a subset of $\mathbb{R}^{\mathbb{N}}$, and each such update descends continuously on compact subdomains on which its denominator is bounded away from zero.*

Proof. Map π to the vector of all its test expectations. Its fibres are exactly (13.2). Any statistic determining the expectations has fibres contained in these fibres, so the vector factors through that statistic as a set map. For the update, the numerator of a future prediction is a one-step pulled-back test and the denominator is the pulled-back constant. Equivalent states have equal numerators and denominators, hence equal normalized predictions.

In the countable continuous case use $q(\pi) = (\pi(v_j))_{j \geq 1}$. After scaling the tests to be bounded by one, its range is a compact subset of $[-1, 1]^{\mathbb{N}}$, with its product metric. A continuous map from a compact space into a Hausdorff space is a quotient map. The fibre-respecting continuous update therefore factors continuously on the stated compact subdomain. No assertion of a Borel cross-section, or of a smooth quotient for arbitrary Borel tests, is used. The minimality statement in the first paragraph alone would not imply either assertion. $\square$

The proposition is relative to the prepared Bayesian experiment. An identity for one prior is not a parameter-uniform comparison of the family $\{K^\lambda\}_\lambda$. All experiment distances below explicitly take the supremum over the parameter when that stronger meaning is intended.

13.3. Attainable Bayes geometry. Let U be square integrable in a separable Hilbert space H , let $\mathcal{H}$ be the actual information, and put $m = \mathbb{E}[U \mid \mathcal{H}]$. Define

$$e_M^2(m) = \inf_{z_1, \dots, z_M \in H} \mathbb{E} \min_i \|m - z_i\|^2.$$

Independent encoder/decoder randomization is permitted, but a retained observation is not a free input after encoding.

Proposition 13.2 (Checkpoint geometry and information loss). *The optimum squared loss of a single M -label checkpoint statistic is*

$$\mathbb{E}\|U - m\|^2 + e_M^2(m). \quad (13.3)$$

For any sub- σ -field $\mathcal{C} \subset \mathcal{H}$, posterior laws of λ satisfy the barycentre identity $\pi_{\mathcal{C}} = \mathbb{E}[\pi_{\mathcal{H}} \mid \mathcal{C}]$. Consequently convex posterior functionals decrease under coarsening. Whenever the quantities are finite,

$$I(\lambda; \mathcal{H}) - I(\lambda; \mathcal{C}) = \mathbb{E} D_{\text{KL}}(\pi_{\mathcal{H}} \parallel \pi_{\mathcal{C}}). \quad (13.4)$$

In a differentiable dominated family, the score of a statistic S is $\mathbb{E}[s_\lambda \mid S]$, and the lost Fisher information matrix is $\mathbb{E} \text{Cov}(s_\lambda \mid S)$, provided differentiation and conditioning are justified in L^2 .

Proof. Conditional orthogonality gives $\mathbb{E}\|U - z\|^2 = \mathbb{E}\|U - m\|^2 + \mathbb{E}\|m - z\|^2$ for every $\mathcal{H}$ -measurable output, and also after adding independent randomness. Freeze that randomness. There are at most M decoder outputs, so nearest-centre assignment is no worse than the given encoding. Conversely every finite codebook has a measurable nearest assignment, with deterministic tie-breaking. Taking infima proves (13.3), without assuming attainment of an optimal codebook. Conditional expectation against bounded parameter tests gives the barycentre identity. Conditional Jensen proves the convex assertion. The chain rule for the logarithm of posterior densities gives (13.4); finite partitions followed by monotone approximation give the same assertion under the stated finiteness condition. Differentiating the marginal density of S gives the score identity. The matrix law of total variance gives the Fisher identity. These are the classical decision and information identities in [1, 11]; their role here is to specify the geometry being implemented, not to certify recursive closure. $\square$

13.4. Normalized response and observation modules. Let b_θ be a differentiable finite positive measure with $b_\theta(1) > 0$, and let $\pi_\theta = b_\theta/b_\theta(1)$. Differentiating this actual normalization gives

$$\partial_\theta \pi_\theta(f) = \frac{\partial_\theta b_\theta(f) - \pi_\theta(f) \partial_\theta b_\theta(1)}{b_\theta(1)}. \quad (13.5)$$

If the derivative has score density s relative to b_θ , the right side is $\text{Cov}_{\pi_\theta}(f, s)$. Thus a response direction is invisible exactly when its centered score annihilates the linear span of the designated tests. If the observation protocol consists of products from two declared linear test spaces V, W , its null space is $(VW)^\perp$, with $VW = \text{span}\{vw : v \in V, w \in W\}$. This last statement follows immediately from linearity, but it requires that those products really are the observed probes. It does not identify an algebraic multiplicity with a physical noise variance. Formula (13.5) and this annihilator are the normalization and observation-module inputs used to type a contact experiment.

14. CAUSAL MORPHISMS AND CHARGED APPROXIMATION

A resource signature specifies ordered coordinates, for example persistent cardinality, numerical precision, transient workspace, program description, preparation count and physical time. A budget transformer is a monotone map between two such signatures, accompanied by an actual implementation. A coordinate may explicitly be unlimited. The persistent-cardinality results below leave fixed programs, fixed real constants, time and transient computation unlimited unless a more specific implementation is given. There is no claim that these resources are equivalent.

A causal simulation from an experiment $\mathcal{E}$ to $\mathcal{F}$ consists of a source controller, a causal transducer and its initialized state. For every target policy the joint construction must produce the target observation/cost history within a declared parameter-uniform total-variation error. The transducer never sees the unknown parameter or the unobserved physical state. Resource bounds are imposed on this joint construction, not just on the terminal observation kernel. We use $\|P - Q\|_{\text{TV}} = \sup_A |P(A) - Q(A)|$.

Theorem 14.1 (Typed simulation and recursive approximation). *The following statements hold.*

- (i) *Serial causal simulations of errors ϵ_1, ϵ_2 and budget transformers ρ_1, ρ_2 compose with error at most $\epsilon_1 + \epsilon_2$ and transformer $\rho_1 \circ \rho_2$. A simulator with K persistent states and a target observer with M states can be run with KM persistent states. For losses in $[0, L]$, the risk changes by at most $L(\epsilon_1 + \epsilon_2)$. Other resource coordinates are transformed according to the supplied implementations. The identity simulation has zero error and the identity transformer. Independent parallel simulations have additive TV error and the product register, with preparations and other costs charged by their joint implementation.*

(ii) Suppose the true predictive state lies in a metric space (S, d) and updates by $s_t = F_t(s_{t-1}, y_t, a_t)$. For all reports and actions suppose F_t is ρ -Lipschitz in its state, $\rho < 1$. Let an implementable approximate update $\tilde{F}_t$ have local error at most η_t , and let $Q_t : S \rightarrow C_t$ be a measurable finite quantizer with radius ϵ_t . The register

$$\hat{s}_t = Q_t \tilde{F}_t(\hat{s}_{t-1}, y_t, a_t) \quad (14.1)$$

has at most $|C_t|$ states at time t and, on every common observation path,

$$d(s_t, \hat{s}_t) \leq \rho^t d(s_0, \hat{s}_0) + \sum_{j=1}^t \rho^{t-j} (\epsilon_j + \eta_j) =: b_t. \quad (14.2)$$

A data-dependent persistent numerical residue must be part of C_t ; it cannot be placed in an uncharged side register. If the report kernels are L_0 -Lipschitz in total variation as functions of the predictive state, the length- n simulated history differs by at most $L_0 \sum_{t \leq n} \mathbb{E} b_t$. The same statement holds up to a bounded stopping time, with the stopped sum. If the true conditional mean is $g(s_t)$ with g L_1 -Lipschitz into a Hilbert space, its implemented prediction has Bayes excess at most $L_1^2 \mathbb{E} b_t^2$.

Proof. Compose the two transducers on their joint initialized state. Applying a Markov kernel contracts total variation, so the triangle inequality gives the sum of errors, uniformly over the policy and parameter for which the two certificates were supplied. The joint register is the Cartesian product. This proves the cardinality bound, and integration of a loss with range $[0, L]$ proves the risk bound. The identity is the unchanged history. For independent parallel laws, insert one product factor at a time and use TV contraction to obtain the same sum bound. The composition order of the budget maps follows from first realizing the target in the intermediate experiment and then realizing that implementation in the source.

For (ii), insert $F_t(\hat{s}_{t-1}, y_t, a_t)$ between the two updates. Contraction, the local defect and the quantization radius give $d(s_t, \hat{s}_t) \leq \rho d(s_{t-1}, \hat{s}_{t-1}) + \eta_t + \epsilon_t$. Induction proves (14.2). For the law comparison, telescope the products of the history kernels, replacing one kernel at a time. The prefix before the replaced kernel is distributed according to the true experiment; its integrated local TV error is at most $L_0 \mathbb{E} b_t$. All suffix kernels are contractions in TV. Summing proves the bound. An absorbing stopped state proves the stopped version by the identical calculation. The policy is evaluated on the same history in each local comparison: no continuity of a possibly discontinuous policy is presumed. Finally apply conditional orthogonality as in Proposition 13.2 and the Lipschitz bound for g . $\square$

Corollary 14.2 (Resource transport of an attainable profile). *If an M -state target observer has risk R_M and a K -state causal simulation has error ϵ for its bounded decision problem, then the source has a KM -state observer of risk at most $R_M + L\epsilon$. Two simulations give the corresponding two directed inequalities; they need not preserve the same cardinality. A K -valued decoder-side event can always be absorbed into a register at a factor K , but a continuous side observation admits no such finite-factor assertion in general.*

Proof. Use the product construction. For the finite event store its current value as the second register coordinate. For the last assertion, an exact uniform real observation supplied directly to a decoder has zero squared estimation error with one otherwise empty state. If it is available only before an M -label encoding, scalar quantization has error $1/(12M^2) > 0$. Thus no finite factor simulates that side interface for all accuracies. $\square$

The theorem joins quotient approximation to experiment comparison. Contractivity is a property to be verified on the true quotient, and Q_t is an executable representation of it. A finite-dimensional formula for the conditional mean is sufficient for one application, but is not a hypothesis of the

theorem. In Section 17 the true state is an entire probability measure. In the next section a different structure replaces metric contraction by an exact finite-tree update.

15. A STRUCTURAL THEOREM FOR PREDICTIVE TREES

15.1. Local hypotheses and the finite profile. Let $\mathcal{L}$ be a pruned factorial language on a finite alphabet of size $b \geq 2$. Thus prefixes and suffixes of legal words are legal, every word has an infinite continuation, and only legal children occur in its rooted tree. Write $[v]$ for a cylinder in the boundary Ω and $\emptyset$ for the empty word. Assume that along any branch there are at most U consecutive vertices with only one legal successor. This condition allows unary vertices but excludes an infinite deterministic branch. Let μ be a stationary probability on Ω , positive on every legal cylinder, and put $p_v = \mu[v]$.

A predictive energy is a positive family a_v , with $a_\emptyset = 1$. Put $D_v = (a_v/p_v)^{1/2}$. The following are the hypotheses of the structural theorem. There are constants $a_- > 0$, $\rho < 1$, and $c_0, C_0 > 0$ such that

$$a_- a_v \leq a_{vi} \leq \rho a_v, \quad \sum_{i: vi \in \mathcal{L}} a_{vi} \leq \rho a_v, \quad (15.1)$$

$$a_{uv} < a_v \quad (u \neq \emptyset), \quad D_{vi} \leq D_v. \quad (15.2)$$

These statements concern legal words only. A bounded Borel map Z from Ω into a separable Hilbert space satisfies

$$\text{diam } Z([v]) \leq C_0 D_v, \quad (15.3)$$

$$\|Z(\omega) - Z(\omega')\| \geq c_0 D_u \quad \text{if } u \text{ is their longest common prefix.} \quad (15.4)$$

The image may be infinite dimensional. Separation is an explicit geometric hypothesis, not a consequence of the word notation.

Starting with the root, repeatedly split a leaf of maximal energy into *all* its legal children. Fix a tie rule. Let $\mathcal{T}_N$ be the last tree in this sequence with at most N leaves, $\mathcal{P}_N$ its leaf cut, and

$$G_N = \sum_{v \in \mathcal{P}_N} a_v. \quad (15.5)$$

A split at a unary vertex is an actual step. Finiteness of $\mathcal{T}_N$ will follow below. The construction is a mathematical finite algorithm; computable implementation of arbitrary real energies is an additional input, not part of their Borel specification.

There are two causal interpretations of these data.

Deletion model. Independent acquisitions $\omega \sim \mu$ occur at renewal times of independent positive integer length L , with $\mathbb{E}L < \infty$. The acquired sequence is observed exactly. At age k the desired output is $g(\sigma^k \omega)$, where g is bounded and Hilbert valued. Write

$$w_k = \mathbb{P}(L > k)/\mathbb{E}L, \quad W(n) = \sum_{k \geq n} w_k, \quad Z(\omega) = (\sqrt{w_k} g(\sigma^k \omega))_{k \geq 0}.$$

In addition to (15.1)–(15.4), assume the local oscillation bound

$$\sum_{k < |v|} w_k (\text{osc}_{[v_k:]} g)^2 + 4 \|g\|_\infty^2 W(|v|) \leq C_1 D_v^2. \quad (15.6)$$

A register sees the renewal event only during its update; the decoder receives neither that event nor a stored raw observation for free.

Insertion model. Let $(Y_t)_{t \in \mathbb{Z}}$ be a stationary finite-alphabet observed process whose reversed past $\omega_t = (Y_t, Y_{t-1}, \dots)$ has law μ and language $\mathcal{L}$. For a square-integrable stationary Hilbert target U_t , suppose

$$m_t = \mathbb{E}[U_t \mid Y_s : s \leq t] = Z(\omega_t), \quad B = \mathbb{E}\|U_t - m_t\|^2. \quad (15.7)$$

The finite register may start at zero with a fixed state; it is not initialized with the infinite past. Infinite stationary pasts here specify the reference prediction and its law. A finite warm-up is included in the limiting average. No renewal or independent-observation assumption is imposed in this interpretation.

For either model let R_M be the infimum, over all M -value registers with causal standard Borel update and decoding kernels, of the upper limiting expected average squared loss. Absolute time and independent coding randomization are permitted. In the insertion model put $E_M = R_M - B$.

Theorem 15.1 (Predictive-tree realization in two orientations). *Under (15.1)–(15.4),*

$$cG_M \leq e_M^2(Z) \leq CG_M. \quad (15.8)$$

Every complete greedy tree is suffix closed. A tree with L leaves has exactly $1 + \sum_{v \text{ internal}} d^+(v)$ vertices, and at most $(U+1)(2L-1)$ vertices; all of them may be used as designated states. For every fixed $B_0 \geq 1$, $G_{\lceil B_0 M \rceil} \asymp_{B_0} G_M$.

In the deletion model, with (15.6),

$$cG_M \leq e_M^2(Z) \leq R_M \leq CG_M. \quad (15.9)$$

In the insertion model,

$$cG_M \leq e_M^2(Z) \leq E_M \leq CG_M. \quad (15.10)$$

The converses apply to the lower limiting average of each randomized, time-dependent observer. The upper bounds are achieved in order by stationary deterministic updates. In particular a logarithmic age or history multiplier is not required. Constants depend on the displayed structural hypotheses, not on the register budget.

15.2. Balance, dilation and exact closure.

Lemma 15.2 (Greedy balance). *For all sufficiently large N , $N/b \leq |\mathcal{P}_N| \leq N$. If θ_N is the largest energy of a current leaf, then*

$$a_- \theta_N \leq a_v \leq \theta_N \quad (v \in \mathcal{P}_N). \quad (15.11)$$

The trees are finite and their profiles have the fixed-dilation property in Theorem 15.1.

Proof. Suppress unary vertices in any finite complete tree. A rooted tree with L leaves and no unary internal vertex has at most $2L-1$ vertices, since its number of edges is the sum of its internal outdegrees. Each vertex of the suppressed tree accounts for itself and at most U unary vertices on its incoming path, including the initial root path. Thus the original tree has at most $(U+1)(2L-1)$ vertices. An infinite sequence of splits with bounded leaf number is impossible, proving finiteness. The split following $\mathcal{T}_N$ crosses the budget and adds at most $b-1$ leaves. Hence $|\mathcal{P}_N| > N - b + 1$.

The maximal leaf energy never increases. Every present leaf was born with energy at least a_- times an earlier maximal energy. This proves (15.11), including any unary splits. Set $\ell = U+1$. Every vertex has at least two distinct descendants at depth ℓ , and at least 2^j at depth ℓj . Choose j with $2^j > b(B_0+2)$. If the later maximal energy at budget $\lceil B_0 N \rceil$ were less than $a_-^{\ell j+1} \theta_N$, every original leaf would have been split through all these ℓj levels: all intervening vertices have energy at least that threshold. The resulting number of leaves would exceed $\lceil B_0 N \rceil$, a contradiction. Therefore the later maximum is at least a fixed multiple of θ_N . Applying (15.11) at the later budget gives $G_{\lceil B_0 N \rceil} \geq c_{B_0} N \theta_N \geq c_{B_0} G_N$. Splitting decreases the energy sum by (15.1), giving the reverse inequality. Adjusting constants covers the finite initial range. $\square$

Lemma 15.3 (Two exact operations on one tree). *Every proper suffix of an internal greedy-tree vertex is internal, and every suffix of any vertex is present. Consequently deletion of the first letter stays in the tree. If v is a leaf and iv is legal, then iv has a leaf prefix in the same tree. It can therefore be updated by prepending i and reading only the letters of v until that leaf is reached.*

Proof. Suppose an internal word uv , $u \neq \emptyset$, is about to be split. If its suffix v has not been split, some current leaf t is a prefix of v . Child decrease and strict prefix-attachment decrease give $a_t \geq a_v > a_{uv}$, contrary to maximality. Thus the suffix is already internal. For a leaf $i_1 \cdots i_n$ with $n \geq 2$, its parent is internal; the suffix $i_2 \cdots i_{n-1}$ of that parent is internal, so its legal child $i_2 \cdots i_n$ is present. The one-letter case ends at the root. Iteration gives closure of every vertex.

If there were no leaf prefix of iv , the complete tree would contain iv as an internal vertex. Its suffix v would then be internal by the first assertion, contradicting that it is a leaf. This proves the insertion assertion. Only legal children were used in either proof. $\square$

15.3. Static lower bound and the causal converse.

Lemma 15.4 (Arbitrary centres). *The bounds (15.8) hold for arbitrary Hilbert-space centres, not only centres on $Z(\Omega)$.*

Proof. Choose one image point in each leaf cylinder to obtain $e_N^2(Z) \leq C_0^2 G_N$. For the lower bound use $\mathcal{P}_{4bM}$, which has at least $4M$ leaves for large M . The open tubes about $Z([v])$ of radii δD_v are disjoint for a sufficiently small fixed $\delta > 0$. Indeed for incomparable v, v' their common prefix u has $D_u \geq \max(D_v, D_{v'})$, and (15.4) separates the two images by at least $c_0 D_u$. Take $2\delta < c_0$.

At most M of the tubes contain one of M proposed centres. On every unoccupied cylinder the squared error is at least $\delta^2 D_v^2$. Its integrated contribution is $\delta^2 a_v$. By balance, the at least three-quarters unoccupied leaves carry at least $3a_-/4$ of the total energy sum. Thus $e_M^2(Z) \geq cG_{4bM} \geq c'G_M$ by Lemma 15.2. The finite initial range is harmless: (15.4) and positive cylinder masses at unbounded branching depths give infinitely supported image law, so every fixed finite-centre error is positive. Alternatively the same tube proof at a larger fixed cut proves that positivity directly. $\square$

Lemma 15.5 (Renewal converse). *In the deletion experiment every randomized, nonstationary M -state observer has lower limiting average loss at least $e_M^2(Z)$.*

Proof. A standard Borel probability kernel is realized as a measurable function of its input and an independent uniform random variable [2]. Applying this at the countably many update and decode times gives an independent coding sequence. A data-dependent persistent position in such a sequence, if used, belongs to the charged state; the representation itself supplies no history channel.

Condition on an acquisition at a fixed time, its pre-acquisition history, and the independent coding variables. The acquired ω still has law μ . Along a hypothetical continuation with no new acquisition, each of the at most M post-acquisition labels determines one output string. For a truncation at age n , the survival-weighted loss is therefore at least $\mathbb{E}L e_M^2(Z^{<n})$, where $Z^{<n}$ is the first n coordinates of the weighted orbit. At a fixed physical time, events specifying the last acquisition are disjoint. Summing this bound over acquisition times at least n before the end of a horizon and using the Cesàro renewal count gives lower limiting average at least $e_M^2(Z^{<n})$. For completeness, the elementary renewal theorem gives expected renewals per unit time tending to $1/\mathbb{E}L$; a fixed truncation removes at most n terminal time indices, each with bounded contribution. No stationary coding rule is presumed.

Finally $e_M^2(Z^{<n}) \uparrow e_M^2(Z)$. Monotonicity follows by projection. The reverse limiting inequality follows by appending zero tails to an arbitrarily good truncated code: the extra mean squared norm is at most $\|g\|_\infty^2 W(n) \rightarrow 0$. This completes the proof. $\square$

Proof of Theorem 15.1. The static assertion, finite count, dilation and suffix closure are proved above. In the deletion experiment retain every tree vertex as a designated state. At acquisition store the unique leaf prefix of ω . Between acquisitions delete its first letter until the empty word is reached, and then keep that word. Choose a representative in each nonempty cylinder and output

its g value; use a fixed bounded output at the empty word. For an acquired leaf v the weighted error is bounded by the left side of (15.6). Averaging and applying renewal reward gives at most $C_1 G_N$. Lemma 15.5 gives the unrestricted converse.

For insertion, after reaching a leaf use the prepend-and-parse rule of Lemma 15.3. During initialization store the observed reversed word while it is internal, or its first leaf prefix when one appears. This uses only vertices of the same tree and after at most its maximum depth gives the correct leaf prefix of the full observed past. Decode a representative Z value in that leaf cylinder. Its stationary excess loss is at most $C_0^2 G_N$ by (15.3) and conditional orthogonality. The finite warm-up has zero limiting average contribution. The converse holds at each time: freeze independent coding randomness, apply orthogonality against m_t , and use the at most M decoder values to lower bound excess by $e_M^2(Z)$. Stationarity makes this the same number at every time, even for time-dependent decoders.

In either orientation choose N a fixed sufficiently small multiple of M , so that $(U+1)(2|\mathcal{P}_N|-1) \leq M$. Dilation converts G_N into $O(G_M)$ without a time-depth multiplier. One fixed state handles the remaining bounded range of budgets. This proves the two risk assertions. $\square$

The static part of the proof uses familiar separated-cylinder quantization ideas [9, 6]. Maximal-energy prefix splitting is likewise a classical coding device [7]. The additional conclusion is that a strict *predictive* energy inequality makes the adaptive tree executable in both orientations, with its whole persistent state counted.

16. TWO MODEL VERIFICATIONS

16.1. Markov-renewal orbit geometry. Let A be a primitive zero-one matrix on b symbols, with Perron root greater than one. Its legal one-sided sequences have a stationary fully supported law μ such that every legal conditional child probability is at least $\zeta > 0$. Let h_i map a compact set of diameter at most one into pairwise g -separated subsets of itself. Assume word scales r_v , $r_\emptyset = 1$, satisfy

$$\begin{aligned} D_0^{-1} r_v |x - y| &\leq |h_v x - h_v y| \leq r_v |x - y|, \\ D_0^{-1} r_u r_v &\leq r_{uv} \leq r_u r_v, \quad 0 < r_- \leq r_i \leq r_+ < 1. \end{aligned} \quad (16.1)$$

The injective coding map identifies the legal sequences with a compact set K , on which T deletes the first symbol. Let $f : K \rightarrow \mathbb{R}^m$ obey

$$|f(x) - f(y)| \leq H|x - y|^\beta, \quad \sum_{j=0}^J |f(T^j x) - f(T^j y)|^2 \geq c_o |x - y|^{2\beta}, \quad 0 < \beta \leq 1. \quad (16.2)$$

The renewal age weights satisfy $0 < q_- \leq w_{k+1}/w_k \leq q_+ < 1$. Set $\tau = 2\beta$ and

$$V(v) = \sum_{k < |v|} w_k r_{v_k}^\tau + W(|v|), \quad a_v = \mu[v]V(v), \quad Z(x) = (\sqrt{w_k} f(T^k x))_{k \geq 0}. \quad (16.3)$$

Corollary 16.1 (Markov renewal as a structural instance). *The data (16.1)–(16.3) satisfy Theorem 15.1 in its deletion orientation. In particular $e_M^2(Z) \asymp R_M \asymp G_M$, with a stationary finite-tree observer and an unrestricted randomized, nonstationary converse. Forbidden edges are not completed, and unary vertices are charged.*

Proof. Every suffix scale is at most one and appending a letter decreases its scale. If $|u| = m$ and $|v| = n$, monotonicity of w gives

$$\begin{aligned} V(v) - \left(\sum_{j < n} w_{m+j} r_{v_j}^\tau + W(m+n) \right) &\geq r_v^\tau \sum_{k < m} w_k, \\ V(uv) - \left(\sum_{j < n} w_{m+j} r_{v_j}^\tau + W(m+n) \right) &\leq r_+^\tau r_v^\tau \sum_{k < m} w_k. \end{aligned}$$

Therefore $V(uv) < V(v)$. Stationarity gives only $\mu[uv] \leq \mu[v]$, which is enough to give the strict energy inequality. Equality of cylinder masses at a unique predecessor causes no problem.

Appending a legal letter multiplies old spatial terms by a factor between $D_0^{-\tau} r_-^\tau$ and r_+^τ . The old tail $W(n)$ becomes $w_n r_i^\tau + W(n+1)$. Since $w_n \geq (1 - q_+)W(n)$ and $W(n+1) \geq q_-W(n)$, we have

$$\ell V(v) \leq V(vi) \leq \rho V(v), \quad \ell = \min(D_0^{-\tau} r_-^\tau, q_-), \quad \rho = 1 - (1 - r_+^\tau)(1 - q_+) < 1.$$

Use $a_- = \zeta \ell$ and sum the conditional masses to verify (15.1). These inequalities also give monotonicity of $D_v = \sqrt{V(v)}$. In a primitive graph with Perron root greater than one, a deterministic path cannot run for b repeated unary vertices: a repeated vertex would generate a closed deterministic cycle. Thus one can take $U = b$.

Introduce $d_\beta(x, y)^2 = \sum_k w_k |T^k x - T^k y|^{2\beta}$. Hölder continuity bounds the orbit distance from above by Hd_β . Summing the delayed lower bound (16.2), and using $w_{k-j} \leq q_-^j w_k$, bounds d_β from above by a constant times the orbit distance. Points in one cylinder have squared d_β distance at most $V(v)$. Points whose longest common prefix is u have, before its end, distances at least $D_0^{-1} g r_{u_k}$, and at its end distance at least g . The last term controls $W(|u|)$ since $W(n) \leq w_n/(1 - q_+)$. This proves both image comparisons. Finally the Hölder bound on each suffix and the bounded tail prove (15.6). The structural theorem now supplies all combinatorial, quantization, and causal conclusions. $\square$

This corollary recovers the earlier Markov-renewal profile without repeating its tree argument as a new model-specific proof. Gibbs pressure identifies the profile exponent under the additional thermodynamic assumptions of the previous Markov theorem, whose full proof is retained in the companion. Pressure is a computation of this instance, not an assumption of the foundational realization theorem. For clarity, the passage from power bounds to an exponent uses only the following elementary fact.

Proposition 16.2 (Logarithmic squeeze). *If for every $\delta > 0$ a positive sequence R_M satisfies $c_\delta M^{-\alpha-\delta} \leq R_M \leq C_\delta M^{-\alpha+\delta}$ eventually, then $-\log R_M / \log M \rightarrow \alpha$.*

Proof. Taking logarithms gives lower and upper limits between $\alpha - \delta$ and $\alpha + \delta$, because fixed logarithmic constants divided by $\log M$ vanish. Let δ decrease to zero. In a pressure-root proof one first chooses the auxiliary root parameter on either side of its zero and only then lets M tend to infinity; root continuity then supplies the arbitrary δ used here. $\square$

16.2. An infinite hidden shift with noisy continual reports. Let $(B_t)_{t \in \mathbb{Z}}$ be independent identically distributed Bernoulli variables. The hidden state at time t is the whole sequence $X_t = (B_t, B_{t-1}, \dots)$. A binary memoryless channel reports Y_t about B_t . Write

$$\gamma = \mathbb{P}(Y_t = 1) \in (0, 1), \quad a_i = \mathbb{P}(B_t = 1 \mid Y_t = i), \quad a_0 \neq a_1.$$

Fix $0 < r < 1$. The Hilbert-valued prediction target and its mean are

$$U_t = (\sqrt{1 - r^2} r^j B_{t-j})_{j \geq 0}, \quad m_t = (\sqrt{1 - r^2} r^j a_{Y_{t-j}})_{j \geq 0}. \quad (16.4)$$

This squared loss can equally be implemented by drawing a fresh terminal query j with probability $(1 - r^2)r^{2j}$ and scoring the prediction of B_{t-j} . The independent query is a task index, not a new observation of the hidden state. Its distribution and interface are fixed in advance.

Theorem 16.3 (Sharp register law for an infinite posterior). *Put $p_0 = 1 - \gamma$, $p_1 = \gamma$ and let $s \in (0, 1)$ be the unique solution of*

$$r^{2s}(p_0^s + p_1^s) = 1, \quad \alpha = (1 - s)/s. \quad (16.5)$$

For the actual noisy shift experiment,

$$R_M = B + \Theta((a_1 - a_0)^2 M^{-\alpha}), \quad B = \sum_{i=0}^1 p_i a_i (1 - a_i). \quad (16.6)$$

The constants in the excess comparison depend on r, p_0, p_1 , not on a_0, a_1 . No posterior truncation is assumed in deriving this law. For a fair hidden bit and a binary symmetric channel of error $\varepsilon < 1/2$, it becomes

$$R_M = \varepsilon(1 - \varepsilon) + \Theta((1 - 2\varepsilon)^2 M^{2 \log r / \log 2}).$$

At $\varepsilon = 1/2$ the predictive mean is constant and the excess is zero with one state.

Proof. The full posterior is the product of Bernoulli laws with parameters $a_{Y_{t-j}}$. Thus (16.4) is the actual conditional mean, and orthogonality gives the displayed B . Normalize its nonconstant part by $a_1 - a_0$. On the full binary past language take $p_v = \prod p_{v_j}$, $D_v = r^{|v|}$ and $a_v = p_v r^{2|v|}$. Children multiply energy by $p_i r^2$, their sum multiplies it by r^2 , and every nonempty prefix attachment strictly decreases it. If two pasts first differ at position n , their normalized squared image distance lies between $(1 - r^2)r^{2n}$ and r^{2n} . All hypotheses of Theorem 15.1 hold in the insertion orientation.

For the exact order of G_M , set $q_i = (p_i r^2)^s$. Equation (16.5) says $q_0 + q_1 = 1$. For every finite complete cut, consistency of this Bernoulli measure gives $\sum_v a_v^s = 1$. By greedy balance, for a cut with L leaves and maximum θ , $L(a_- \theta)^s \leq 1 \leq L\theta^s$. Hence $G_M \asymp L\theta \asymp L^{-(1-s)/s} \asymp M^{-\alpha}$. The root exists uniquely because the left side of (16.5) decreases continuously from two to $r^2 < 1$. The fair case solves it explicitly, and the constant-mean case is immediate. If only reports from time zero are supplied, the unobserved past contributes at most $r^{2(t+1)}$ to the squared target norm at time t . Its average vanishes; the finite-tree initialization in the structural theorem still gives (16.6). $\square$

This is not an exactly closed finite list of conditional means. Every past report changes its own posterior coordinate, and arbitrarily many coordinates can be queried. It also has no fixed-step common minorizer: after k hidden updates the old tail remains intact, and two different initial tails give disjoint supports. A measure dominated by all those kernels must be zero. As always, a bare Borel encoding of an infinite sequence into one real number is possible; it is not a finite-cardinality register and does not change the theorem.

17. FINITE PRESENTATIONS OF MEASURE-VALUED FILTERS

The preceding sharp example has a separated tree geometry. We next give a constructive approximation theorem requiring no such separation and no finite-dimensional exact posterior coordinate.

Lemma 17.1 (A finite Wasserstein presentation). *For $n \geq 1$ let $\mathcal{C}_n$ consist of empirical probability measures with n equally weighted atoms on $\{0, 1/n, \dots, 1\}$. Then*

$$|\mathcal{C}_n| = \binom{2n}{n} \leq 4^n, \quad \sup_{\pi \in \mathcal{P}([0,1])} W_1(\pi, Q_n \pi) \leq \frac{3}{2n} \quad (17.1)$$

for a specified Borel quantizer Q_n .

Proof. Take the generalized quantiles of π at $(j - 1/2)/n$, $1 \leq j \leq n$, and round them to nearest grid points, breaking ties toward zero. Quantiles are Borel functions of a probability law. If q is its monotone quantile function, the integral of the error from replacing q by its midpoint value on each of the n equal subintervals is at most $1/n$: bound each integral by its interval length times the oscillation and sum the monotone variations, at most one. Rounding adds at most $1/(2n)$. In one dimension W_1 is the L^1 distance of quantile functions, giving the bound. The possible sorted lists are multisets of size n from $n + 1$ points, counted by $\binom{2n}{n}$. The binomial theorem bounds this by 4^n . $\square$

Theorem 17.2 (A charged measure-valued recursive approximation). *Let the actual filter obey $\pi_t = \Phi_{y_t, a_t}(\pi_{t-1})$ on $\mathcal{P}([0, 1])$, and suppose every allowed update is ρ -contractive in W_1 , with $\rho < 1$. Suppose an implementable update has W_1 defect at most η . The initialized register*

$$\hat{\pi}_t = Q_n \tilde{\Phi}_{y_t, a_t}(\hat{\pi}_{t-1}), \quad \hat{\pi}_0 \in \mathcal{C}_n$$

has at most $\binom{2n}{n}$ persistent values and satisfies, for every actual observation path,

$$W_1(\pi_t, \hat{\pi}_t) \leq \rho^t + \frac{3/(2n) + \eta}{1 - \rho}. \quad (17.2)$$

This bounds the error of every 1-Lipschitz test expectation, uniformly over the choice of test. Thus for $M \geq 4$, $n = \lfloor \log M / \log 4 \rfloor$ gives error $O((\log M)^{-1} + \eta)$ and squared error $O((\log M)^{-2} + \eta^2)$ after the initialization term. If these test expectations are conditional means, the latter is a Bayes excess bound. Report-law comparison additionally requires the report Lipschitz hypothesis of Theorem 14.1; it is not automatic from W_1 contraction.

Proof. Apply Theorem 14.1 with the true metric state $S = \mathcal{P}([0, 1])$, radius $3/(2n)$, and diameter one. The dual formula for W_1 gives the assertion for every test; independent selection of a terminal test does not enlarge its bound. The codebook count and the logarithmic substitution are exact. Only an index of a codebook law persists between updates, not its exact unquantized posterior or a numerical residual. $\square$

17.1. A nonlinear physical realization. Let B_t be independent fair bits and define

$$h_0(x) = x/5 + x^2/50, \quad h_1(x) = 3/5 + x/5 + x^2/50, \quad X_t = h_{B_t}(X_{t-1}). \quad (17.3)$$

The observation Y_t reports B_t correctly with probability $3/4$, independently at each time. A permitted terminal probe $g : [0, 1] \rightarrow [0, 1]$ returns a Bernoulli variable of probability $g(X_t)$; the tested family is 1-Lipschitz probes and their bounded finite executable continuations. The actual update-report instrument on measures is

$$\mathsf{T}_y \pi = \frac{1}{2} \left\{ \frac{3}{4} (h_y)_\# \pi + \frac{1}{4} (h_{1-y})_\# \pi \right\}.$$

Its mass is $1/2$, independent of the old state. The normalized filter is therefore

$$\Phi_y \pi = \frac{3}{4} (h_y)_\# \pi + \frac{1}{4} (h_{1-y})_\# \pi. \quad (17.4)$$

Corollary 17.3 (Nonlinear posterior, explicit register). *The actual measure-valued filter (17.4) is $6/25$ -contractive in W_1 . Theorem 17.2 applies with $\eta = 0$ to an exact rational implementation on $\mathcal{C}_n$. The model has no fixed-step nonzero common minorizer. For every integer $D \geq 1$, the first D polynomial moments do not determine their own exact next-step update on the space of all prior laws.*

Proof. The two images lie respectively in $[0, 11/50]$ and $[3/5, 41/50]$. Each derivative is between $1/5$ and $6/25$. Couple two old laws optimally and use the same branch draw in their mixtures; this gives $W_1(\Phi_y \pi, \Phi_y \pi') \leq (6/25) W_1(\pi, \pi')$. For a codebook input, (17.4) has at most $2n$ atoms, at rational points with denominator dividing $50n^2$, with weights multiples of $1/(4n)$. Sorted cumulative weights give the exact quantiles, which are rounded back to the fixed grid. Thus no

numerical limit is being hidden in $\eta = 0$. The transient list of at most $2n$ atoms is discarded; its entries and weights have $O(\log n)$ -bit rational descriptions. The designated persistent object is the single codebook index. Approximate arithmetic instead obeys the separately charged η bound of the theorem.

After k hidden transitions the law from a fixed x has finite support $\{h_v(x) : |v| = k\}$. For a second initial point x' , each equality $h_v(x') = h_u(x)$ excludes at most one x' , since h_v is strictly increasing. Choose x' outside this finite set. The two supports are disjoint, so a measure dominated by all k -step laws must be zero.

For the moment assertion, the D th updated moment integrates $\frac{3}{4}h_y(x)^D + \frac{1}{4}h_{1-y}(x)^D$, a polynomial of degree $2D$ with nonzero leading coefficient 50^{-D} . It is not in the span of $1, x, \dots, x^D$. Linear algebra on finitely many distinct points therefore supplies a signed measure annihilating this span but not the new polynomial. Its positive and negative parts have equal mass; normalizing them yields two prior probabilities with identical first D moments and different updated D th moment. The argument is about ordinary finite moment closure, not the impossibility of a Borel encoding of measures by real numbers. $\square$

The quotient here is genuinely the posterior law for the specified probe family: bounded Lipschitz functions separate probability measures on a compact interval, and constants do not affect that separation. Updates respect this quotient by (17.4). The advancing reports have the exact state-independent fair law, while a terminal Bernoulli probe has total-variation defect at most $W_1(\pi_t, \hat{\pi}_t)$. Thus the same finite presentation defines a causal experiment simulation with a controlled terminal error and an exactly charged register. Multiple probes would update the posterior again; no claim for an unverified enlarged probe protocol is implicit. This example connects positive instruments, predictive quotient, recursive approximation and resource transformation within one model.

18. SHARP FINITE-EXPERIMENT BUDGETS

For experiments $\mathcal{E} = \{P_\theta\}$ and $\mathcal{F} = \{Q_\theta\}$ with the same parameter set, write

$$\delta(\mathcal{E}, \mathcal{F}) = \inf_K \sup_\theta \|KP_\theta - Q_\theta\|_{\text{TV}}, \quad \Delta(\mathcal{E}, \mathcal{F}) = \max\{\delta(\mathcal{E}, \mathcal{F}), \delta(\mathcal{F}, \mathcal{E})\}.$$

All parameterized laws in this section are Borel families. The infimum is over parameter-independent Markov kernels. These are terminal experiment comparisons in the sense of [11, 12]; only the encoding statistic, not the comparison randomization, is part of the physical data reduction.

Let $\mathcal{G} = \{N(z, \Gamma) : z \in K\}$, where $K \subset \mathbb{R}^r$ is compact and Γ is positive definite. In the lower bound assume that K contains a nonempty Euclidean ball.

Theorem 18.1 (Gaussian finite-experiment cardinality). *There are constants $c, C > 0$, depending on K, Γ, r , such that*

$$cM^{-2/r} \leq \inf_{\mathcal{F}: |\mathcal{F}| \leq M} \Delta(\mathcal{F}, \mathcal{G}) \leq CM^{-2/r}, \quad M \geq 1. \quad (18.1)$$

The lower bound ranges over all experiments on at most M outcomes, not just deterministic Gaussian quantizers. The upper bound is attained in order by a deterministic statistic with J^r labels, where $J = \lfloor M^{1/r} \rfloor$, followed by a specified positive reconstruction kernel. No overflow label is needed.

18.1. The cardinality obstruction.

Lemma 18.2 (A decision-theoretic lower bound). *Under the preceding assumptions, $\Delta(\mathcal{F}, \mathcal{G}) \geq cM^{-2/r}$ whenever $\mathcal{F}$ has at most M outcomes.*

Proof. Restrict the mean parameter to a closed cube K_0 of positive side length contained in K and put the uniform prior on K_0 . Restriction can only decrease an experiment distance. Use squared-error estimation of z , with actions restricted to K_0 ; its loss range is bounded by $L = \text{diam}(K_0)^2$. Let $m(y) = \mathbb{E}[z \mid y]$ in this Gaussian experiment. Differentiation under the compact prior integral gives

$$Dm(y) = \text{Cov}(z \mid y)\Gamma^{-1}. \quad (18.2)$$

The posterior density is positive on the cube. Hence its covariance is positive definite, and $Dm(y)$ is nonsingular for every y . The inverse function theorem on a small neighborhood of any fixed y_0 , together with the positive continuous marginal density of y , shows that the law of $m(y)$ has a density bounded below by a positive constant on some smaller cube in $\mathbb{R}^r$.

This implies $e_M^2(m) \geq c_0 M^{-2/r}$. To see it directly, put balls of radius $t = (|Q|/(2Mv_r))^{1/r}$ about the M proposed centres, where Q is the smaller cube and v_r the volume of the unit ball. At least half of Q remains outside them, and contributes at least a fixed multiple of t^2 . This argument permits arbitrary centres.

Every M -outcome garbling S of the Gaussian observation therefore has Bayes risk at least $B_{\mathcal{G}} + c_0 M^{-2/r}$, by conditional orthogonality. If $\Delta(\mathcal{F}, \mathcal{G}) < \epsilon$, choose a kernel K from $\mathcal{G}$ to $\mathcal{F}$ with error below ϵ . The experiment $K\mathcal{G}$ has at most M outputs, and bounded-loss comparison gives

$$B_{K\mathcal{G}} \leq B_{\mathcal{F}} + L\epsilon \leq B_{\mathcal{G}} + 2L\epsilon.$$

The second inequality uses a kernel in the reverse direction. Comparing these bounds and taking the infimum over ϵ proves the result. Randomized actions do not improve a Bayes squared-error minimum. Notice that $\mathcal{F}$ itself was not assumed to be a Gaussian garbling. $\square$

18.2. Positive second-order reconstruction. Put

$$\tau(x) = \frac{1}{2} + \pi^{-1} \arctan x, \quad u = \tau^{\otimes r}(y) \in (0, 1)^r. \quad (18.3)$$

Let p_z be the transformed Gaussian density. If $x_j = \tan \pi(u_j - 1/2)$, then

$$p_z(u) = \phi_{\Gamma}(x - z) \prod_{j=1}^r \pi(1 + x_j^2). \quad (18.4)$$

It extends by zero across every face to a smooth periodic density. Indeed each fixed-order derivative is a polynomial in x times the Gaussian density, which tends to zero faster than any polynomial as a face is approached. Uniformity holds for z in the fixed compact set. For use below define the finite constants

$$H_2 = \sup_{z, u} \|D^2 p_z(u)\|_{\text{op}}, \quad H_0 = \max_j \sup_{z, u_j} p_{z, j}(u_j), \quad (18.5)$$

where $p_{z, j}$ is a transformed marginal density. These suprema, or any upper bounds obtained from (18.4), fix all the reconstruction and boundary constants; they do not depend on M or sample size. For instance, differentiating twice introduces only polynomials and the eigenvalue bounds of Γ , so bounded mean norm and fixed covariance conditioning suffice.

Partition the flat unit torus into J^r cubes of side $h = 1/J$, with centres c_i , and write $Q_J(u)$ for the cube label. For $J \geq 2$ let Φ_i be the periodic tensor product of the piecewise-linear hat functions with value one at c_i , zero at its neighboring mesh centres, and support within one mesh step in each coordinate. Then

$$\Phi_i \geq 0, \quad \int \Phi_i = h^r, \quad \sum_i \Phi_i(u) = 1.$$

On each mesh interval the neighbouring centres reproduce the affine coordinate, using local periodic lifts across the seam. Define the reconstruction from label i to have density $h^{-r} \Phi_i(u)$, then apply τ^{-1} coordinatewise. Values on the faces have probability zero.

Lemma 18.3 (A quadratic reconstruction estimate). *Uniformly in $z \in K$, this kernel H_J satisfies*

$$\|H_J Q_J \tau^{\otimes r} N(z, \Gamma) - N(z, \Gamma)\|_{\text{TV}} \leq \frac{r H_2}{12 J^2}. \quad (18.6)$$

Proof. Let p_i be the probability of cube i . Symmetric Taylor expansion about its centre gives $|p_i/h^r - p_z(c_i)| \leq r H_2 h^2/24$. The linear interpolant $I_h p_z = \sum_i p_z(c_i) \Phi_i$ has error at most $r H_2 h^2/8$: its weights reproduce constants and affine functions and the sum of their coordinate variances is at most $r h^2/4$. The second Taylor remainder is half this variance times H_2 . Both calculations hold across the seam because p_z has a smooth periodic extension. The reconstructed density is $\sum_i (p_i/h^r) \Phi_i$. Nonnegativity and the partition of unity give supremum error at most $r H_2 h^2/6$. Integrating over volume one and using the factor $1/2$ for total variation proves the claim. Inverting (18.3) preserves this distance. $\square$

Proof of Theorem 18.1. Lemma 18.2 gives the lower bound. The deterministic statistic $Q_J \tau^{\otimes r}$ has J^r outputs, so the forward deficiency is zero; Lemma 18.3 gives the reverse deficiency. For $M \geq 2^r$, $J = \lfloor M^{1/r} \rfloor \geq M^{1/r}/2$. For the remaining finite range use any one-outcome experiment and the trivial bound $\Delta \leq 1$, increasing C . The reconstruction uses positive probabilities; it is not a signed interpolation operator. $\square$

The use of a compactified grid avoids a growing Gaussian overflow radius. The use of hats rather than constant cells cancels the first Taylor term. Both choices are necessary for the stated proof of a logarithm-free quadratic budget rate.

19. A SHARP BUDGET FOR THE PHYSICAL COUNT-TO-CONTACT EXPERIMENT

We now specify the actual local protocol, following the known-mark experiment of A2 [8]. Fix $\lambda_0 \in (0, \infty)^m$, a matrix $B \in \mathbb{R}^{m \times d}$, a compact set $H \subset \mathbb{R}^d$, and a full-row-rank matrix $C \in \mathbb{R}^{r \times m}$. Observe independent counts

$$K_{N,j} \sim \text{Pois}(N \lambda_{0,j} + \sqrt{N}(Bh)_j), \quad h \in H. \quad (19.1)$$

Write $\Lambda = \text{diag } \lambda_0$, $A = CB$, and $\Gamma = C \Lambda C^T$. The contact score and limiting experiment are

$$S_N = C(K_N - N \lambda_0)/\sqrt{N}, \quad \mathcal{G} = \{N(Ah, \Gamma) : h \in H\}.$$

The physical finite statistic is

$$T_{N,J} = Q_J(\tau^{\otimes r}(S_N)), \quad |T_{N,J}| \leq J^r. \quad (19.2)$$

There is no physical jitter, no retained real score, and no omitted overflow event. Every real score belongs to one of the J^r cells of the open compactified cube, with deterministic boundary tie-breaking.

Theorem 19.1 (Joint sample and register law). *For the fixed protocol (19.1), there are $N_0, C_* < \infty$ such that uniformly for $N \geq N_0$ and $J \geq 2$,*

$$\Delta(\mathcal{E}_{N,J}, \mathcal{G}) \leq C_* \left(N^{-1/2} + \frac{J}{\sqrt{N}} + J^{-2} \right), \quad (19.3)$$

where $\mathcal{E}_{N,J} = \{\text{Law}_h(T_{N,J}) : h \in H\}$. If AH contains an r -dimensional ball, every experiment with at most M outputs has distance from $\mathcal{G}$ at least $c_* M^{-2/r}$. Consequently the statistic with $J = \lfloor M^{1/r} \rfloor$ satisfies

$$c_* M^{-2/r} \leq \Delta(\mathcal{E}_{N,J}, \mathcal{G}) \leq C_{*,K} M^{-2/r}, \quad 2^r \leq M \leq K N^{r/6}, \quad (19.4)$$

for every fixed $K < \infty$. In particular $M \asymp N^{r/6}$ gives distance of order $N^{-1/3}$; achieving distance of order $N^{-1/5}$ requires and is achieved by a budget of order $N^{r/10}$. All constants may depend on the fixed matrices, compact parameter set and covariance conditioning. No uniformity at rank degeneration or unknown marks is asserted.

We prove the uniform estimates in five steps, keeping comparison randomization separate from (19.2).

Lemma 19.2 (Jittered Poisson local normal comparison). *There is an absolute $C_P < \infty$ such that for $\mu \geq 1$, independent $U \sim \text{Unif}(-1/2, 1/2)$, and $K \sim \text{Pois}(\mu)$,*

$$\|\text{Law}(K + U) - N(\mu, \mu)\|_{\text{TV}} \leq C_P \mu^{-1/2}. \quad (19.5)$$

Proof. It suffices to consider $\mu \geq 64$, increasing the constant for the remaining range. The jittered density is $p_\mu(k) = e^{-\mu} \mu^k / k!$ on $[k - 1/2, k + 1/2)$, $k \geq 0$. On the central cells $|k - \mu| \leq \mu^{2/3}/4$, put $t = k - \mu$, $x = k + u$, $|u| \leq 1/2$, and $z = (x - \mu)/\sqrt{\mu}$. Stirling's formula with remainder $0 < R_k < 1/(12k)$ gives

$$\log \frac{p_\mu(k)}{\phi_\mu(x)} = -\frac{1}{2} \log(k/\mu) - R_k - \{k \log(k/\mu) - k + \mu\} + \frac{(x - \mu)^2}{2\mu},$$

where ϕ_μ is the $N(\mu, \mu)$ density. For $|t| \leq \mu/2$, Taylor's theorem applied to $(1 + v) \log(1 + v) - (1 + v) + 1$ gives remainder at most $|t|^3/\mu^2$ after its quadratic term. Also $|\log(1 + t/\mu)| \leq 2|t|/\mu$ and $|(x - \mu)^2 - t^2| \leq |t| + 1/4$. Substitution therefore gives the convenient bound

$$\left| \log \frac{p_\mu(k)}{\phi_\mu(x)} \right| \leq 32\mu^{-1/2}(1 + |z|^3), \quad \left| \log \frac{p_\mu(k)}{\phi_\mu(x)} \right| \leq 64. \quad (19.6)$$

The constants are deliberately loose. For example $|z| \leq \mu^{1/6}/4 + 1/(2\sqrt{\mu})$ on these cells verifies the second bound directly. The inequality $|e^v - 1| \leq e^{|v|}|v|$ now bounds the integrated central density error by $32e^{64}\mu^{-1/2}(1 + \mathbb{E}|G|^3)$, $G \sim N(0, 1)$.

For completeness the omitted tails are exponentially smaller. For $|s| \leq 1$, the centered Poisson log moment generating function is $\mu(e^s - 1 - s) \leq \mu s^2$. Exponential Markov with $s = \mu^{-1/3}/16$, and its negative, bounds the probability outside the central cells, allowing the half-unit jitter margin, by $2 \exp(-\mu^{1/3}/128)$ after adjusting an absolute factor for $64 \leq \mu$ in the bounded initial interval. The Gaussian bound follows by its moment generating function. Thus an absolute choice such as $C_P = 10^8 e^{64}$ dominates the central error, the tails and the finite initial range. This proves a uniform $\mu^{-1/2}$ bound rather than an unquantified local-limit assertion. $\square$

Lemma 19.3 (Variance replacement and projected comparison). *Let U_j be independent uniform half-cell jitters, used only in this proof, and set $S_N^o = S_N + CU/\sqrt{N}$. Uniformly for $h \in H$,*

$$\|\text{Law}_h(S_N^o) - N(Ah, \Gamma)\|_{\text{TV}} \leq C_0 N^{-1/2}. \quad (19.7)$$

One may choose C_0 as a function only of m , the minimum of λ_0 , $\max_{h \in H} \|Bh\|$, and fixed matrix norms.

Proof. Put $b_* = \max_{h \in H} \|Bh\|_\infty$ and $\lambda_* = \min_j \lambda_{0,j}$. Choose $N_0 \geq \max(128/\lambda_*, 4b_*^2/\lambda_*^2)$. Then each count mean is at least $N\lambda_*/2$. Product comparison using Lemma 19.2, followed by centering and scaling, gives distance at most $mC_P\sqrt{2/\lambda_*} N^{-1/2}$ from the normal vector with mean Bh and covariance $\Lambda + N^{-1/2} \text{diag}(Bh)$.

For the replacement of this covariance by Λ , the relative variance eigenvalues are $1 + t_j$, with $|t_j| \leq b_*/(\lambda_*\sqrt{N}) \leq 1/2$. The exact Gaussian relative entropy is $\frac{1}{2} \sum_j \{t_j - \log(1 + t_j)\} \leq \sum_j t_j^2$. The elementary entropy-variation inequality $\|P - Q\|_{\text{TV}} \leq \sqrt{D_{\text{KL}}(P\|Q)/2}$ gives a further bound $\sqrt{m/2} b_*/(\lambda_*\sqrt{N})$. Taking $C_0 = mC_P\sqrt{2/\lambda_*} + \sqrt{m/2} b_*/\lambda_*$ is sufficient. Projection by C contracts total variation and sends the fixed Gaussian covariance to Γ , proving the assertion. $\square$

Lemma 19.4 (Deterministic cell crossing). *Let H_0 be the transformed marginal bound in (18.5), now for the means AH , and put $d_N = \|C\|_{\infty,1}/(2\pi\sqrt{N})$. Then*

$$\sup_{h \in H} \|\text{Law}_h(T_{N,J}) - Q_J \tau^{\otimes r} N(Ah, \Gamma)\|_{\text{TV}} \leq 2C_0 N^{-1/2} + 2rH_0 J d_N. \quad (19.8)$$

Here $\|C\|_{\infty,1}$ is the maximum absolute row sum.

Proof. The coupled scores satisfy $\|S_N - S_N^o\|_{\infty} \leq \|C\|_{\infty,1}/(2\sqrt{N})$. Since $|\tau'| \leq 1/\pi$, their compactified coordinates differ by at most d_N . Distinct grid labels require a coordinate of the jittered score to be within d_N of a grid boundary. On the torus there are J boundaries per coordinate; under the transformed Gaussian their total probability is at most $2rH_0 J d_N$, or its truncation by one. Replacing the jittered law by that Gaussian adds at most $C_0 N^{-1/2}$, by (19.7). The coupling inequality then compares theunjittered and jittered labels. One more data-processing comparison of the jittered labels to the Gaussian labels adds the second $C_0 N^{-1/2}$. Boundary tie events are already in the strips. This argument never asserts that the unjittered real score has a Gaussian law in total variation. $\square$

Proof of Theorem 19.1. Use the deterministic Gaussian-to-cell kernel for the direction $\mathcal{G} \rightarrow \mathcal{E}_{N,J}$. Lemma 19.4 bounds its error by $b_{N,J} = 2C_0 N^{-1/2} + 2rH_0 J d_N$. For the reverse direction, use the positive kernel H_J from Lemma 18.3. Kernel contraction and the triangle inequality give

$$\sup_h \|H_J \text{Law}_h(T_{N,J}) - N(Ah, \Gamma)\|_{\text{TV}} \leq b_{N,J} + rH_2/(12J^2).$$

This proves (19.3), with, for example, $C_* = 2C_0 + rH_0\|C\|_{\infty,1}/\pi + rH_2/12$ increased to one. Every kernel is parameter independent. Jitter was introduced only to bound a difference of laws, not to modify the physical statistic.

If AH contains a ball, restrict to an affine parameter section over a smaller mean cube and apply Lemma 18.2. More generally one may simply index the Gaussian family by its mean and choose a Borel representative preimage for each mean in the cube (the compact fibres of the continuous map on H admit a measurable section); the lower comparison uses only that subfamily. For $J = \lfloor M^{1/r} \rfloor$ and $M \geq 2^r$, $J^{-2} \leq 4M^{-2/r}$. If $M \leq KN^{r/6}$, then $M^{3/r}/\sqrt{N} \leq K^{3/r}$; hence $J/\sqrt{N}$ and $N^{-1/2}$ are also at most fixed multiples of $M^{-2/r}$. This proves the window. Taking M respectively of order $N^{r/6}$ and $N^{r/10}$ gives the two rates. Conversely, an error at most $CN^{-1/5}$ forces $M \geq cN^{r/10}$ by the universal cardinality lower bound. $\square$

19.1. The A2 matrices and the scope of the consequence. In A2's known-mark local protocol, the moment derivative D has $\Sigma = D\Lambda D^T > 0$, and its calibrated exposure direction is $B = \Lambda D^T \Sigma^{-1}$, so $DB = I$. If L maps to the contact coordinates and F is an orthonormal frame of its range, the matrices above are

$$C = F^T L D, \quad A = CB = F^T L, \quad \Gamma = C\Lambda C^T > 0.$$

Here A has full row rank r . A compact local parameter neighborhood therefore has an image containing a mean ball. Both sides of Theorem 19.1 apply to that actual protocol. This invokes a structural experiment-cardinality theorem verbatim, not just an analogy between a root singularity and a contact observation.

The result is a matched budget theorem in the displayed joint window. It is not a lower bound saying that $N^{-1/3}$ is the best sample-size comparison achievable with arbitrarily many labels. It also does not assign a stochastic noise law to a conductor scheme or establish uniformity at a rank-changing physical stratum. The geometric source, the exposure protocol, the finite statistic and the comparison kernels are distinct typed objects throughout.

REFERENCES

- [1] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272. doi:10.1214/aoms/1177729032.
- [2] O. Kallenberg, *Foundations of Modern Probability*, third ed., Probability Theory and Stochastic Modelling, vol. 99, Springer, 2021. doi:10.1007/978-3-030-61871-1.
- [3] T. Linder and S. Yüksel, On optimal zero-delay coding of vector Markov sources, *IEEE Trans. Inform. Theory* **60** (2014), 5975–5991. doi:10.1109/TIT.2014.2346780.
- [4] R. G. Wood, T. Linder and S. Yüksel, Optimal zero delay coding of Markov sources: stationary and finite memory codes, *IEEE Trans. Inform. Theory* **63** (2017), 5968–5980. doi:10.1109/TIT.2017.2692215.
- [5] H. Luschgy and G. Pagès, Sharp asymptotics of the functional quantization problem for Gaussian processes, *Ann. Probab.* **32** (2004), 1574–1599. doi:10.1214/009117904000000324.
- [6] M. K. Roychowdhury, Quantization dimension for Gibbs-like measures on cookie-cutter sets, *Kyoto J. Math.* **54** (2014), 239–257. doi:10.1215/21562261-2642377.
- [7] B. P. Tunstall, *Synthesis of noiseless compression codes*, Ph.D. dissertation, Georgia Institute of Technology, 1967.
- [8] Q. Qi, Conductor strata and nonreduced multiplication failure schemes, A2 revision 118, manuscript, September 22, 2026. Fixed repository edition, commit 44bfc648ead0.
- [9] M. Kesseböhmer and S. Zhu, The upper and lower quantization coefficient for Markov-type measures, *Math. Nachr.* **290** (2017), 827–839. doi:10.1002/mana.201500328. Theorem 1.1 checked in the authors’ preprint, entitled *The quantization for Markov-type measures on a class of ratio-specified graph directed fractals*, arXiv:1406.3257v1.
- [10] Y. E. Demirci, A. D. Kara and S. Yüksel, Average cost optimality of partially observed MDPs: contraction of nonlinear filters and existence of optimal solutions and approximations, *SIAM J. Control Optim.* **62** (2024), 2859–2883. doi:10.1137/24M1643736. Theorem numbering in the text refers to arXiv:2312.14111v3.
- [11] L. Le Cam and G. L. Yang, *Asymptotics in Statistics: Some Basic Concepts*, second edition, Springer, New York, 2000. doi:10.1007/978-1-4612-1166-2.
- [12] E. Torgersen, *Comparison of Statistical Experiments*, Encyclopedia of Mathematics and its Applications 36, Cambridge University Press, Cambridge, 1991.
- [13] C. R. Shalizi and J. P. Crutchfield, Computational mechanics: pattern and prediction, structure and simplicity, *J. Statist. Phys.* **104** (2001), 817–879. arXiv:cond-mat/9907176.
- [14] L. Le Cam, Sufficiency and approximate sufficiency, *Ann. Math. Statist.* **35** (1964), 1419–1455. doi:10.1214/aoms/1177700372.
- [15] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28. doi:10.1524/strm.2002.20.14.1.
- [16] A. Nerode, Linear automaton transformations, *Proc. Amer. Math. Soc.* **9** (1958), 541–544. doi:10.1090/S0002-9939-1958-0135681-9.
- [17] B. R. Kloeckner, A generalization of Hausdorff dimension applied to Hilbert cubes and Wasserstein spaces, *J. Topol. Anal.* **4** (2012), 203–235. doi:10.1142/S1793525312500094. Author preprint: arXiv:1105.0360v2.
- [18] J. Matoušek, *Using the Borsuk–Ulam Theorem: Lectures on Topological Methods in Combinatorics and Geometry*, Universitext, Springer, Berlin, 2003. doi:10.1007/978-3-540-76649-0.
- [19] Q. Qi, Attainable information and causal compression at exponent collisions, A1 revision 37, manuscript, September 9, 2026. Fixed repository edition, commit 90465076589f.
- [20] A. Nayyar, A. Mahajan and D. Teneketzis, Decentralized stochastic control with partial history sharing: a common information approach, *IEEE Trans. Automat. Control* **58** (2013), 1644–1658. doi:10.1109/TAC.2013.2239000. The theorem numbers cited here refer to the author preprint, arXiv:1209.1695.
- [21] H. S. Witsenhausen, On the structure of real-time source coders, *Bell System Tech. J.* **58** (1979), 1437–1451. doi:10.1002/j.1538-7305.1979.tb02263.x.